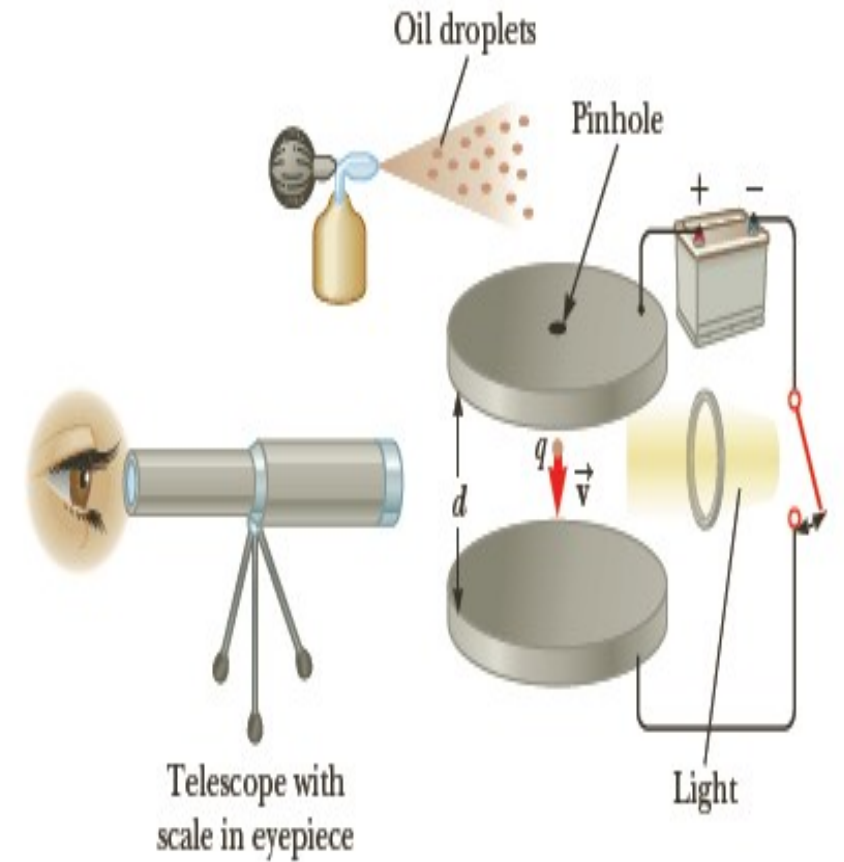


ELECTRIC CHARGES AND FIELDS



**SCHOOL OF
SCIENCE AND
TECHNOLOGY**
PAN-ATLANTIC UNIVERSITY

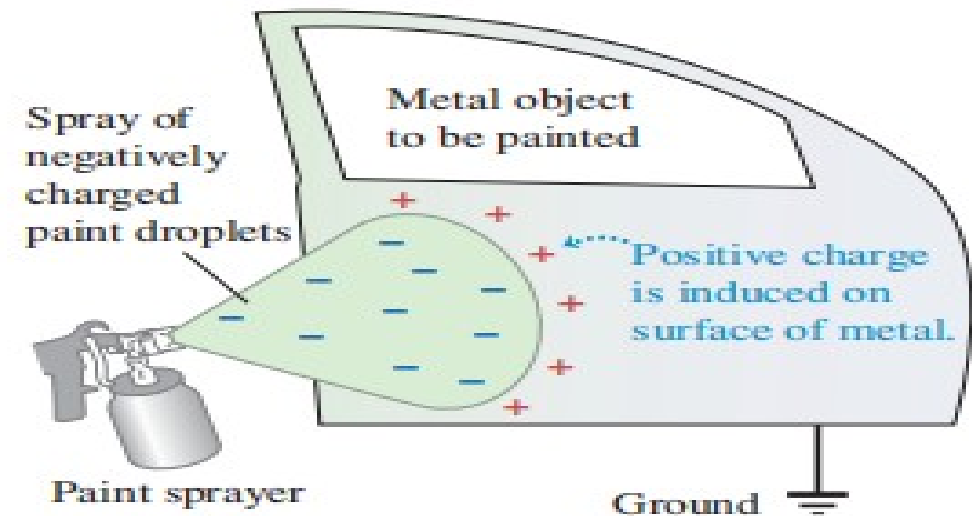
Lecture Overview

1. Electrostatics; conductors and insulators, Electric charge, methods of charging, Properties of electric charge.
2. Coulomb's law; Electric force and Electric field
3. Capacitance and Dielectrics
4. Magnetism and magnetic fields
5. Electromagnetic induction and Maxwell's equations
6. Current electricity; and currents, Ohm's law and analysis of DC circuits
7. AC voltages applied to inductors, capacitors and resistance.

outlines

- Electrostatics;
- Conductors and insulators,
- Electric charge,
- Methods of charging,
- Properties of electric charge.

21.9 The electrostatic painting process (compare Figs. 21.7b and 21.7c). A metal object to be painted is connected to the earth ("ground"), and the paint droplets are given an electric charge as they exit the sprayer nozzle. Induced charges of the opposite sign appear in the object as the droplets approach, just as in Fig. 21.7b, and they attract the droplets to the surface. This process minimizes overspray from clouds of stray paint particles and gives a particularly smooth finish.



ELECTRIC CHARGE

- ☐ What holds your body together?
- ☐ What keeps a skyscraper standing?
- ☐ What holds your car on the road as you round a turn?
- ☐ What governs the electronic circuitry in your computer or smartphone, or provides the tension in your climbing rope?
- ☐ What enables a plant to make sugar from sunlight and simple chemicals?
- ☐ What underlies the awesome beauty of lightning?

The answer, in all cases, is the **electric force**

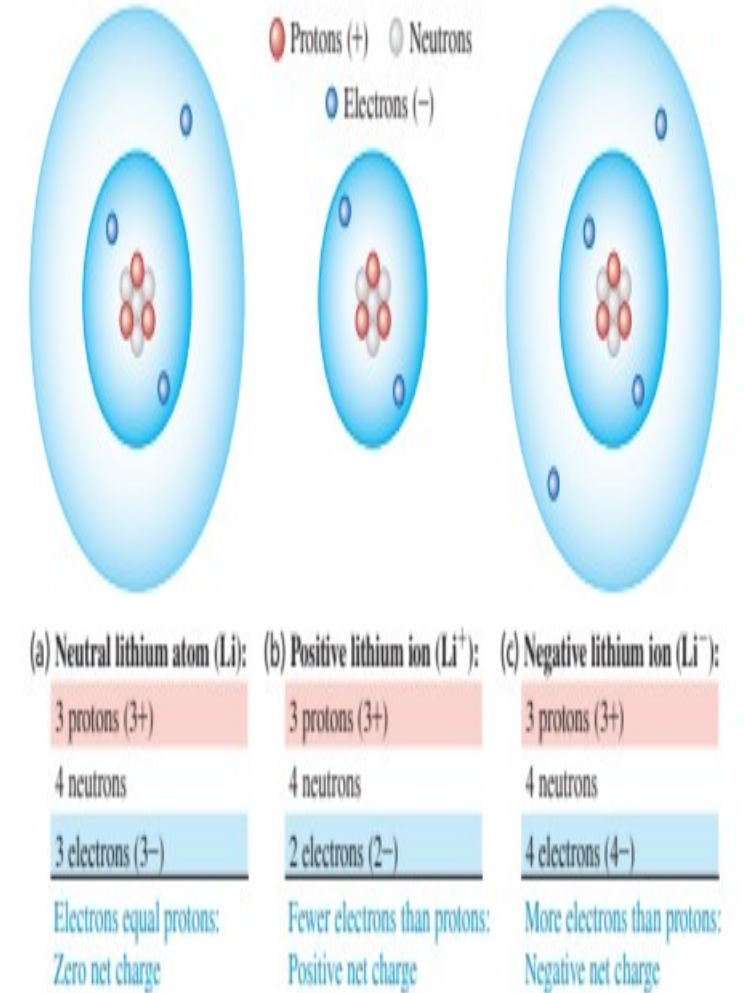
With the exception of gravity, all the forces we've encountered in mechanics—including tension forces, normal forces, compression forces, and friction—are based on electric interactions; so are the forces responsible for all of chemistry and biology.

The electric force, in turn, involves a fundamental property of matter—namely, **electric charge**.

Electric Charge

What is an electric charge?

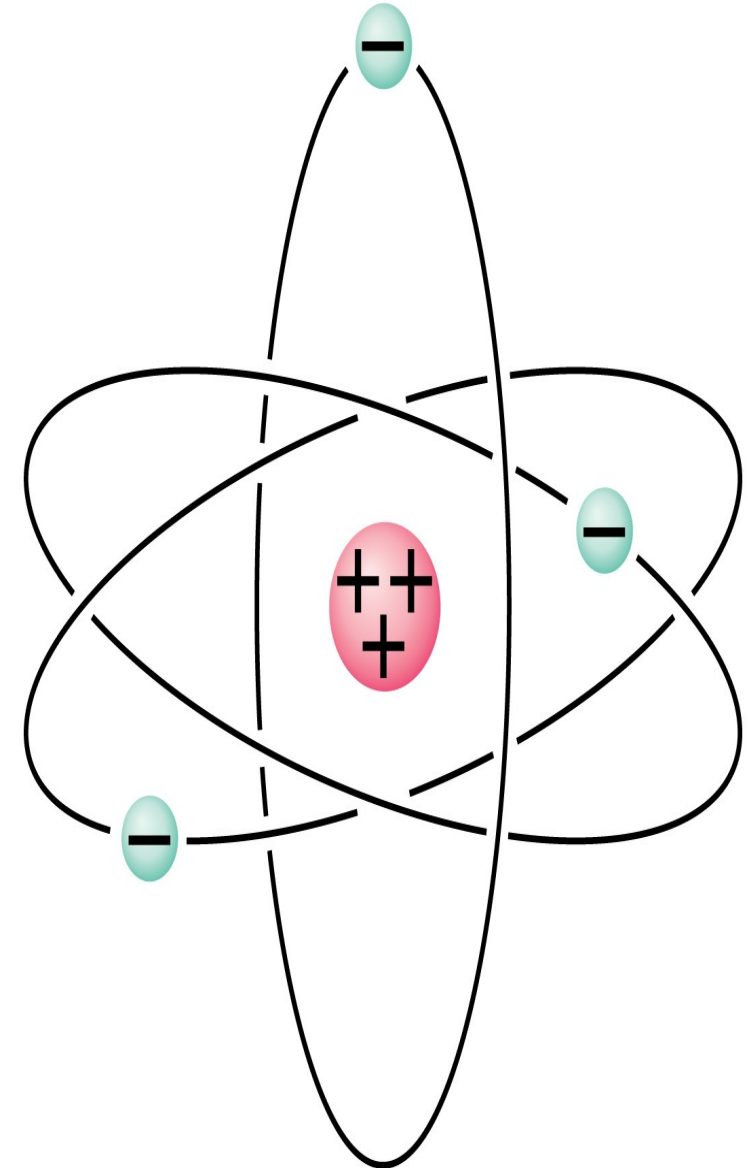
- It is a basic property of matter carried by some elementary particles (electron and proton).
- It is a physical property of matter that causes it to experience a force when kept in an electromagnetic field.
- Charge is divided into two categories: positive and negative **charges**.
- Charge is a property of particles



Electric Charge

Atom Basics

- Atom is said to be electrically neutral because it has equal number of positive protons and negative electrons.
- An Object is said to be *charged* when it has an unequal electrons, of positive/negative charges.
- Only electrons move to create charges.
- When an object gains or loses electrons it becomes CHARGED!!!!
- When atoms lose electrons, they become positively charged ions
- When atoms gains electrons. they become negatively charged ions.



ELECTRIC CHARGE

*The ancient Greeks discovered as early as 600 b.c. that after **they rubbed amber with wool**, the amber could attract other objects. Today we say that the amber has acquired a net electric charge, or has become charged*

The word “**electric**” is derived from the Greek word elektron, **meaning amber**.

What is Electrostatics?

It is the branch of physics that studies the nature of charges that is not moving.

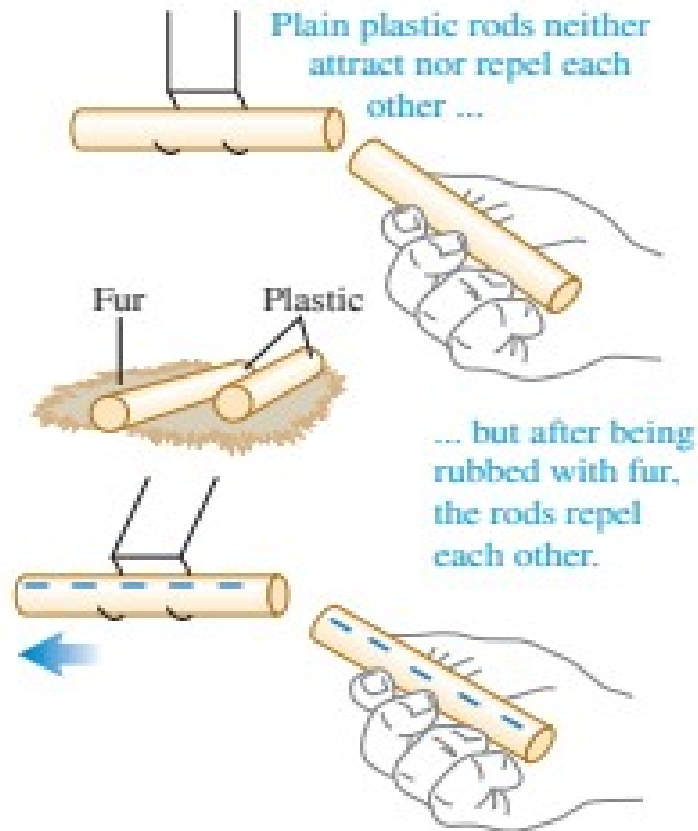
It is the branch of physics that studies electric charges at rest. The study of STATIC ELECTRICITY.

Static electricity is electrical charges that can be collected and held in one place.

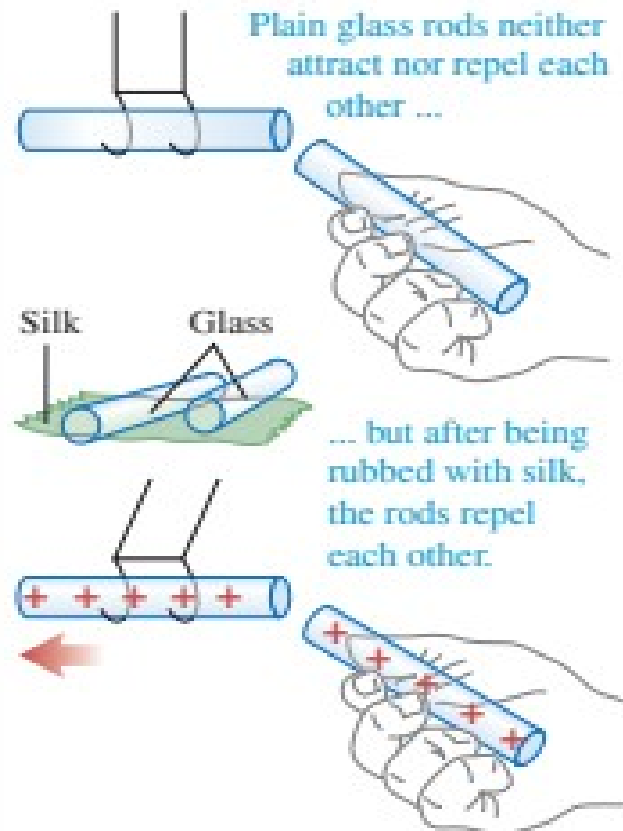
ELECTRIC CHARGE

Figure 21.1 Experiments in electrostatics. (a) Negatively charged objects repel each other. (b) Positively charged objects repel each other. (c) Positively charged objects and negatively charged objects attract each other.

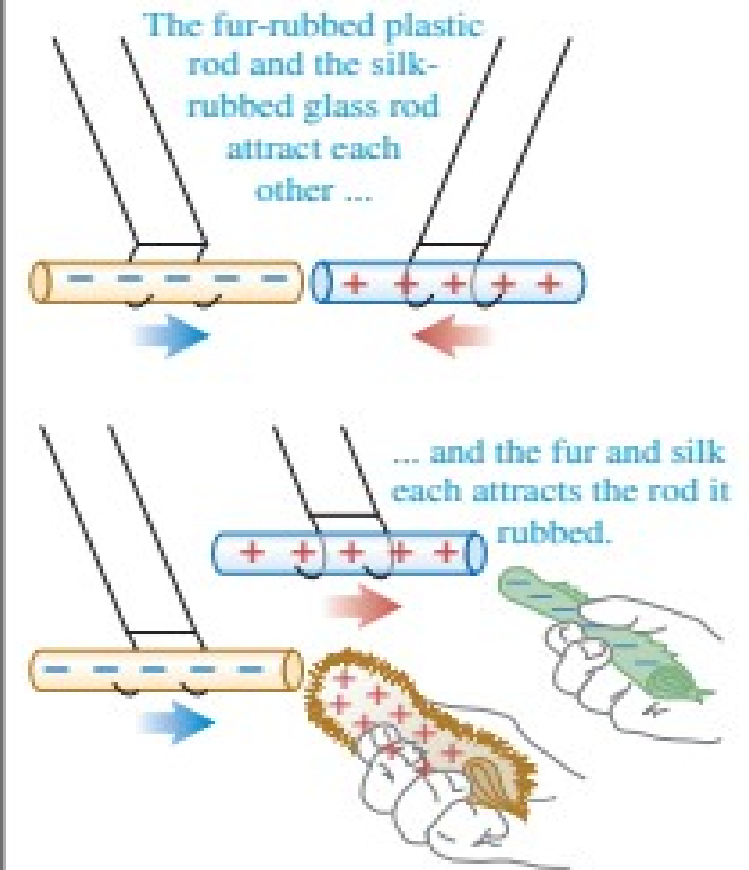
(a) Interaction between plastic rods rubbed on fur



(b) Interaction between glass rods rubbed on silk



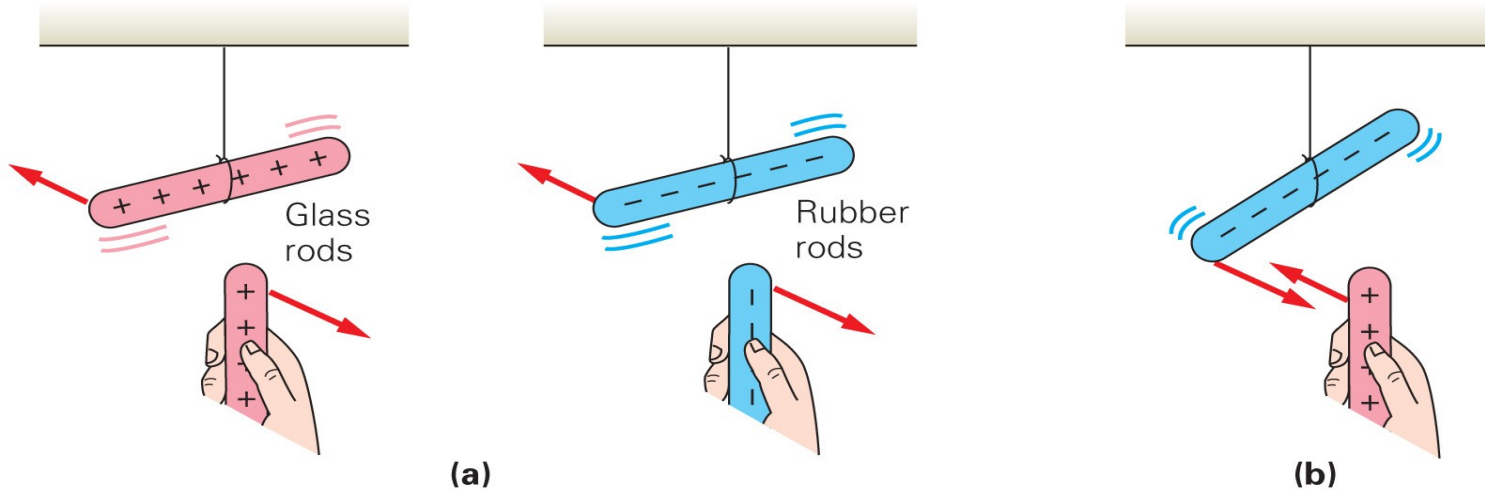
(c) Interaction between objects with opposite charges



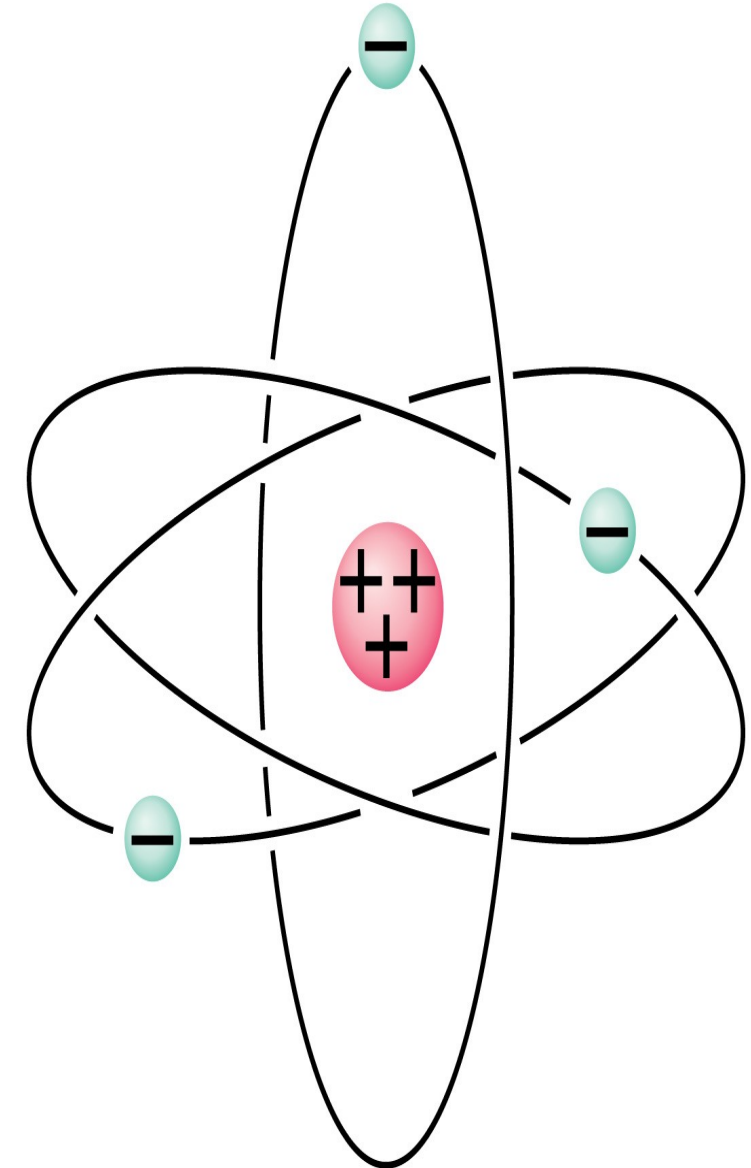
Electric Charge

Fundamental laws of charges

- Unlike charges attract while like charges repel.



- This force of attraction or repulsion between charged particles is called the **ELECTRIC FORCE**
- And the fact that these + or – particles in matter carry electric charge, this is the basic idea of electricity.



Conductors and Insulators

Conductors

- They are materials that permit electrons to flow freely from particle to particle; allows electricity pass through them easily. E.g. copper, iron and steel
- An object made of a conducting material will permit charge to be transferred across the entire surface of the object. If charge is transferred to the object at a given location, that charge is quickly distributed across the entire surface of the object. The distribution of charge is the result of electron movement.

Insulators

- They are materials that impede the free flow of electrons from atom to atom; do not allow electricity to pass through them. E.g. **Plastic, wood, glass and rubber.**
- If charge is transferred to an insulator at a given location, the excess charge will remain at the initial location of charging. The particles of the insulator do not permit the free flow of electrons; subsequently charge is seldom distributed evenly across the surface of an insulator.

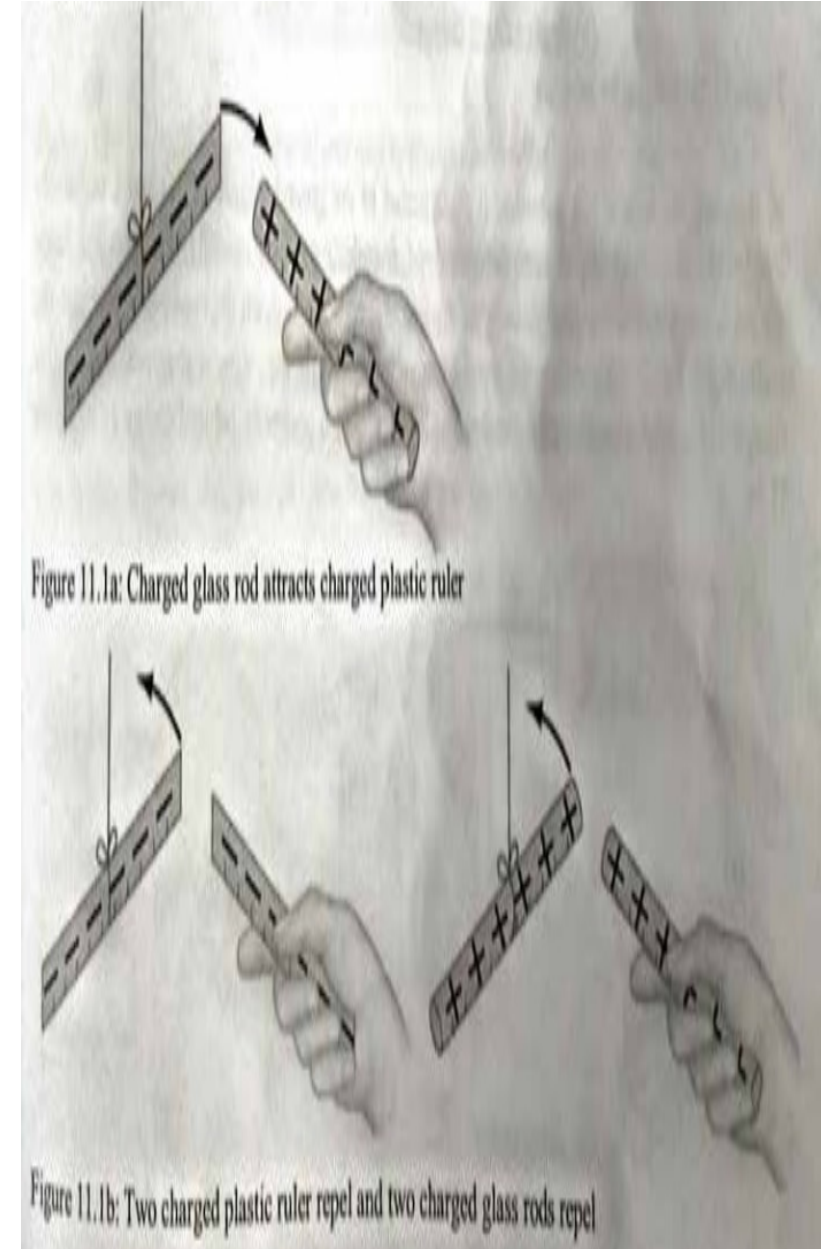
Method of Charging

Charging means gaining or losing electron.

- Material losing electron is positively charged while material gaining electron is negatively charged.
- The number of electrons gained by a material is equal to the number of electrons lost by the other material.

There are three ways a material can be charged:

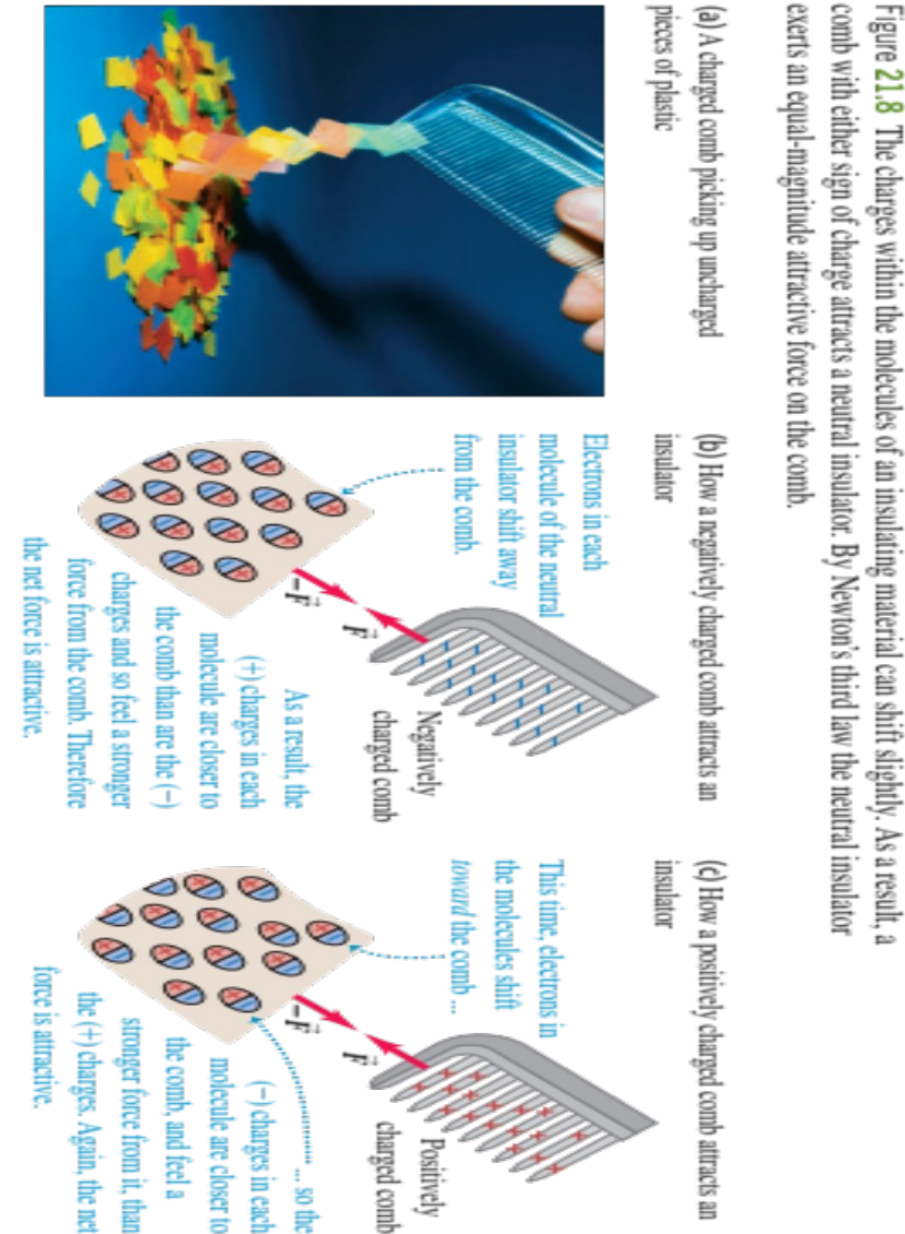
- charging by friction (rubbing)
- charging by contact (conduction) and
- charging by induction.



Method of Charging

Charging by Friction (rubbing)

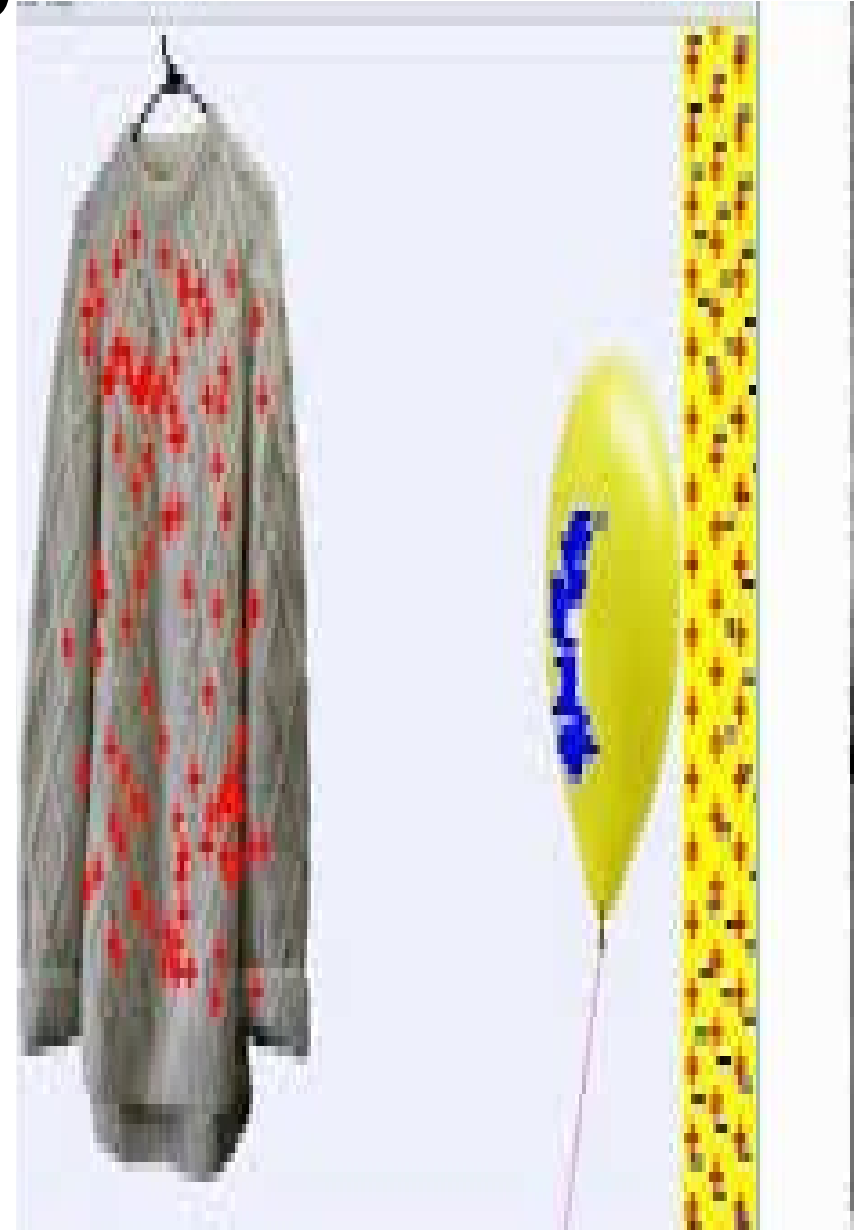
- When two neutral objects are rubbed against each other, one object may pull electrons away from the other creating one positive object and one negative object. This is known as charging by friction. This charging takes place in insulators only. Example:
- When a glass rod is rubbed with a silk cloth, glass acquires a positive charge and silk cloth acquires a negative charge. Electrons are removed from glass rod and are added to silk cloth.
- When an ebonite rod is rubbed with fur cloth, ebonite rod acquires a negative charge and fur cloth acquires a positive charge.



Method of Charging

Charging by Friction

- ❖ Charging in this case are not evenly distributed all over the surface of the material but concentrated at the area where the rubbing took place, this is because in insulator the electrons are not free to move about easily, they are tightly bound.
- ❖ rub balloon on your cloth and move it close to a wall; it sticks to the wall. Electrons on the clothe is transferred to the balloon. The clothe is +ve , the balloon is -ve and the wall neutral.



Method of Charging

Charging by Contact (conduction)

- ❖ Charging by conduction involves the contact of a charged object to a neutral object. Suppose that a positively charged aluminum plate is touched to a neutral metal sphere.
- ❖ The neutral metal sphere becomes charged as the result of being contacted by the charged aluminum plate.

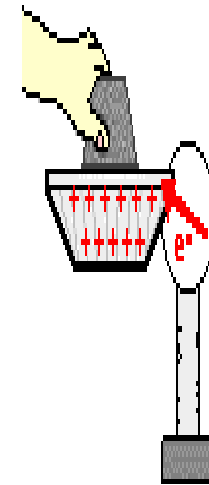
Charging a Neutral Object by Conduction

Diagram i.



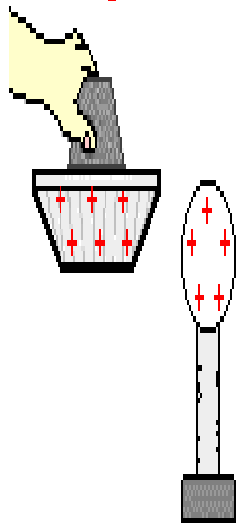
A neutral metal sphere rests upon an insulating platform.

Diagram ii.



When the + aluminum plate is touched to the metal sphere, electrons are drawn off the sphere and onto the aluminum plate.

Diagram iii.



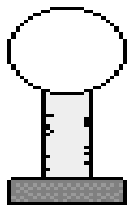
The aluminum plate has less excess + charge and the metal sphere now has an excess of + charge.

Charging by Contact (conduction)

- Charging by conduction involves the contact of a charged object to a neutral object. Suppose that a positively charged aluminum plate is touched to a neutral metal sphere. The neutral metal sphere becomes charged as the result of being contacted by the charged aluminum plate.

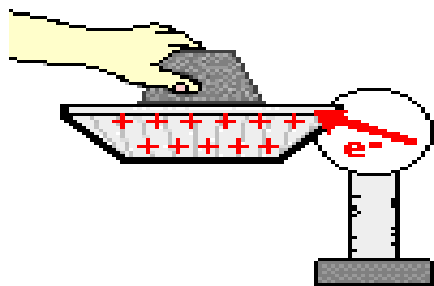
Charging a Neutral Object by Conduction

Diagram i.



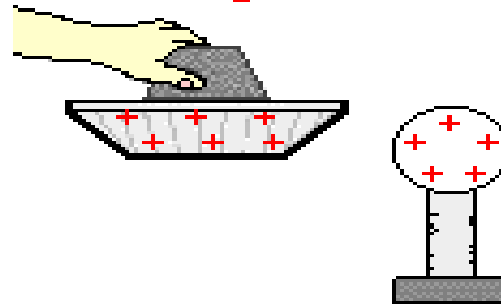
A neutral metal sphere rests upon an insulating platform.

Diagram ii.

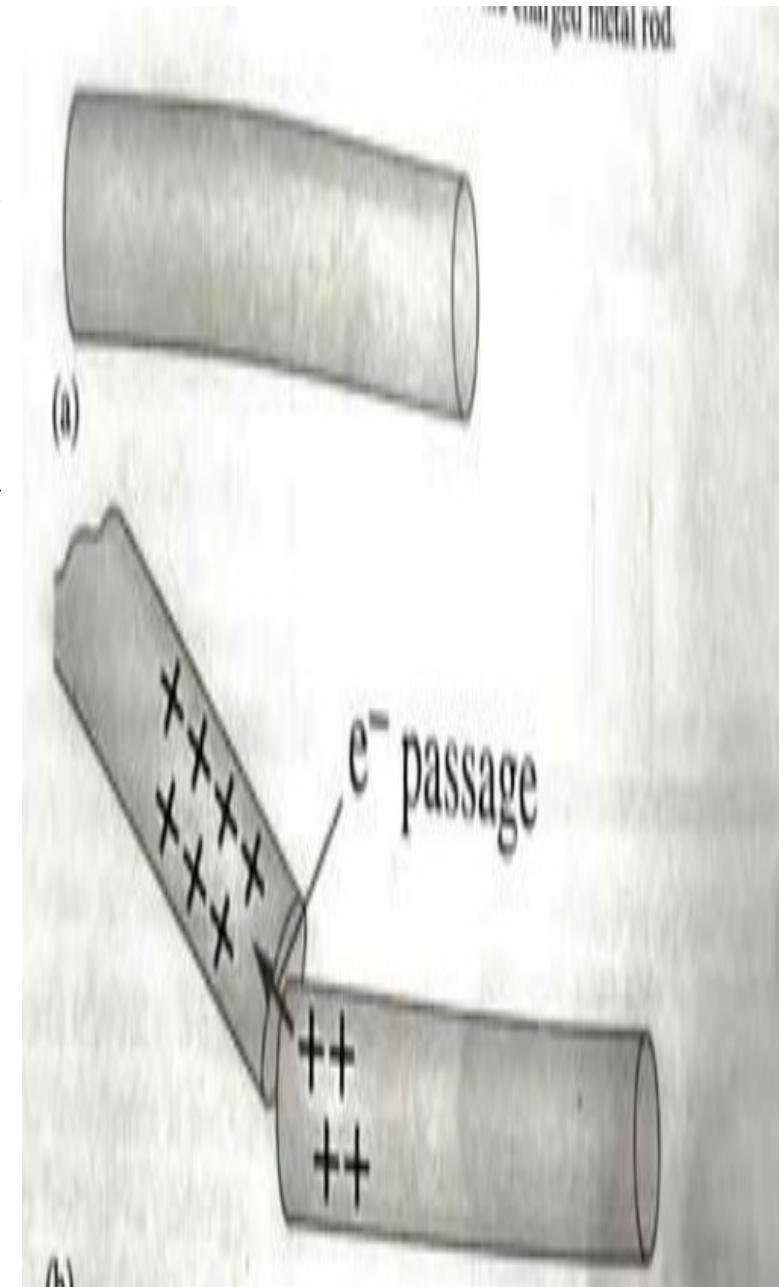


When the + aluminum plate is touched to the metal sphere, electrons are drawn off the sphere and onto the aluminum plate.

Diagram iii.



The aluminum plate has less excess + charge and the metal sphere now has an excess of + charge.

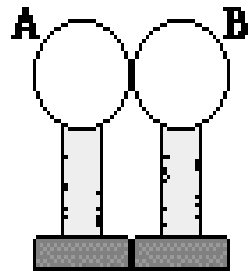


Charging by Induction

Induction charging is a method used to charge an object without actually touching the object to any other charged object.

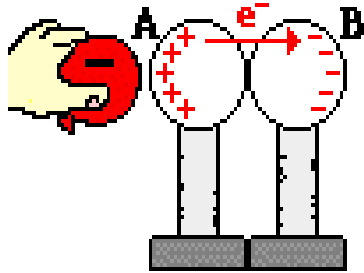
Charging by Induction

Diagram i.



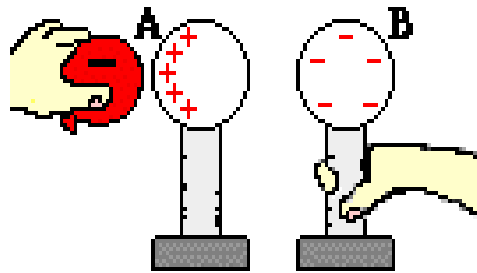
Two metal spheres are mounted on insulating stands.

Diagram ii.



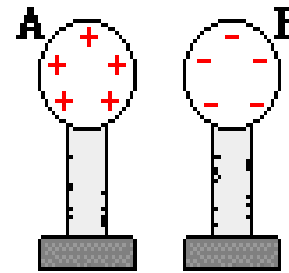
The presence of a - charge induces e^- to move from sphere A to B. The two-sphere system is polarized.

Diagram iii.

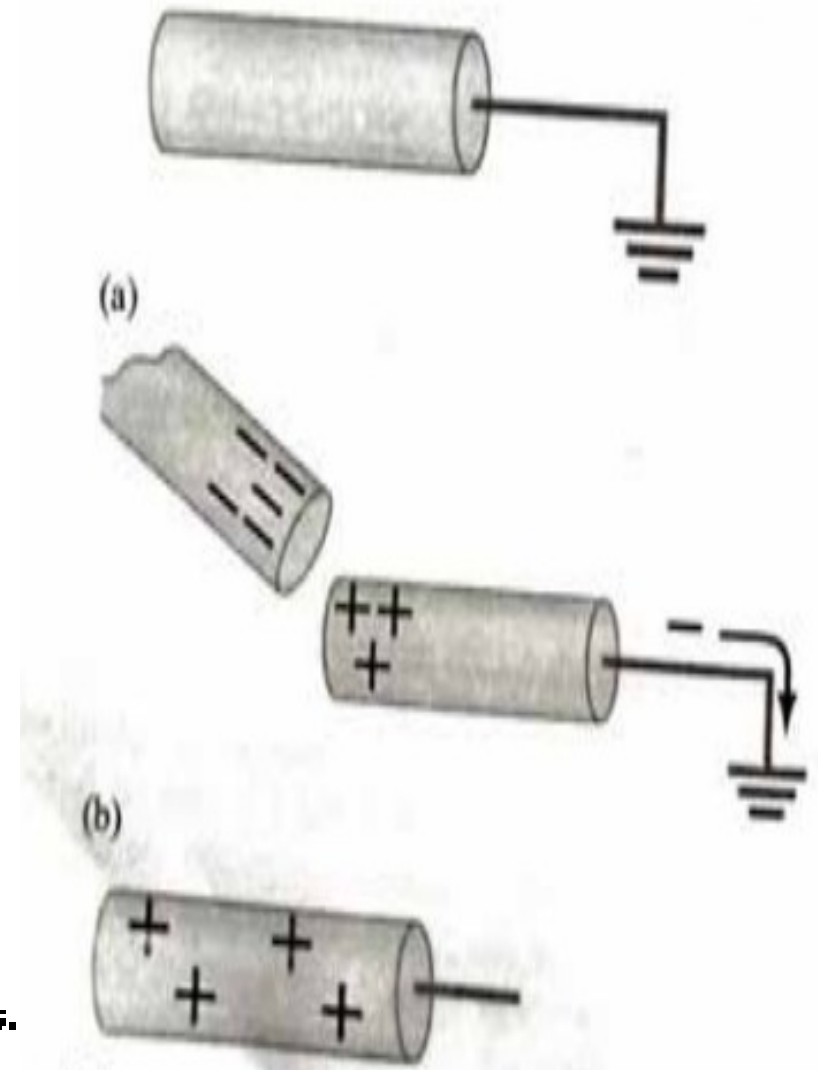


Sphere B is separated from sphere A using the insulating stand. The two spheres have opposite charges.

Diagram iv.

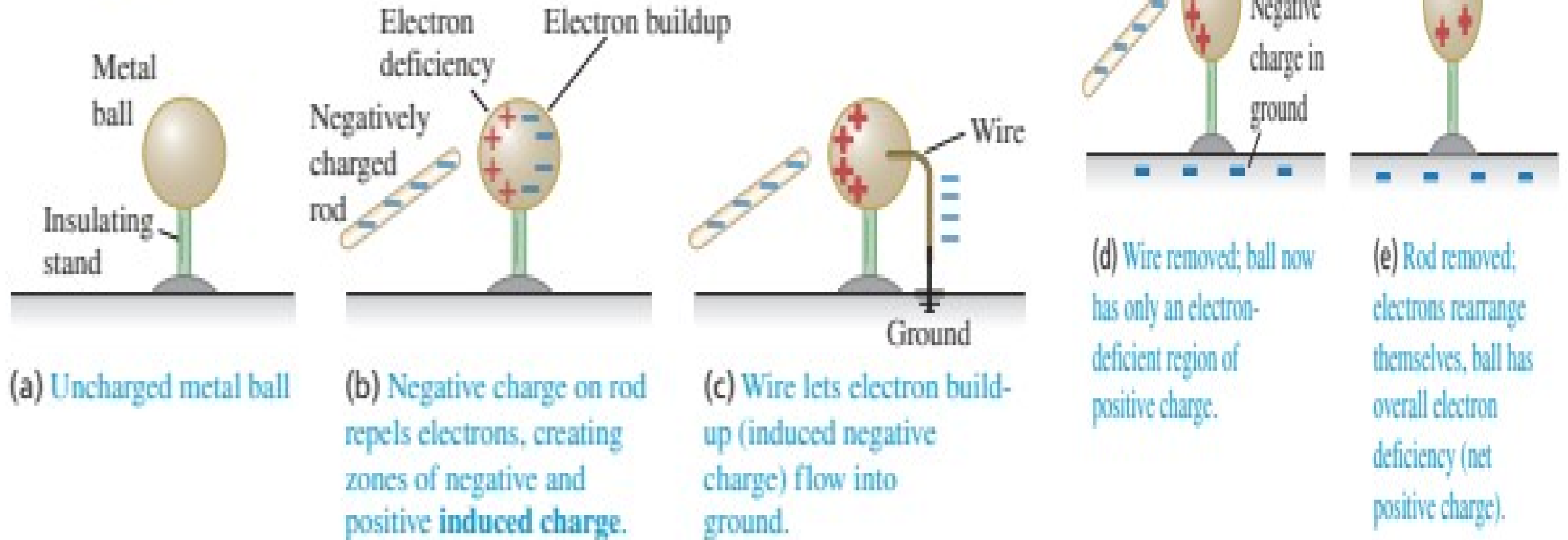


The excess charge distributes itself uniformly over the surface of the spheres.



Charging by Induction

Figure 21.7 Charging a metal ball by induction.



Properties of Electric Charges

- The **charge** is a scalar quantity as it has only magnitude and no direction.
- As charges are of two types, positive and negative, there are other certain basic properties they follow. If the size of charged bodies is so small, we consider them as point charges.

The basic properties of electric charges are as follows:

- Additivity of Charges
- Conservation of Charges
- Quantization of charge

Conservation of Charges

- The charge is a conserved quantity which means charge can neither be created nor be destroyed but can be transferred from one body to another by certain methods like conduction and induction.
- As charging involves rubbing two bodies, it is actually a transfer of electrons from one body to another.
- The number of electrons gained by one body is equal to the number of electrons lost by the other body. Hence, the total number of charges is the same.
- E.g. if 5C is the total charge of the system, then it can be redistributed as 1C , 2C and 2C or in any other possible permutation, but by conservation principle the net charge of system will always be 5C .

Electric Charge and Its Conservation

Electric charge is conserved – the arithmetic sum of the total charge cannot change in any interaction. This is an example of a law that is now well established: *the law of conservation of electric charge, which states that:*

the net amount of electric charge produced in any process is zero;

or, said another way,

no net electric charge can be created or destroyed.

Quantization of charge

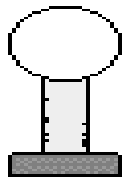
- Quantization of charge means that the charge of any object is an integer multiple of the elementary charge (e – charge on one electron). Suppose charge on a body is q , then it can be written as $q = ne$
- where n is an integer and not fraction or irrational number, like ' n ' can be any positive or negative integer like 1, 2, 3, -5 etc.
- Thus, an object's charge can be exactly $0e$, $1e$, $-1e$, $2e$, etc., but not, say, $1/2 e$, or $-3.8 e$, etc

Earthing or Ground

- In electricity the “ground” is very important. This is generally a huge body that will be electrically neutral.
- The Earth is a great example of “ground.” The human body can also serve as a ground for many cases.
- Earthing is the process of sharing charges with the earth or ground. It provides a safety measure for electrical circuits and appliances.
 - Example: By touching the charged object, you ground it.

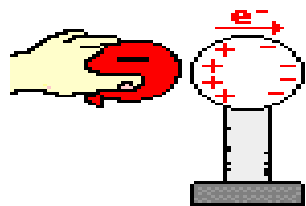
Charging a Single Sphere by Induction

Diagram i.



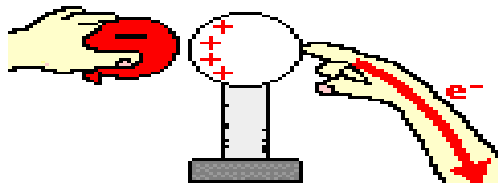
A metal sphere is mounted on a stand.

Diagram ii.



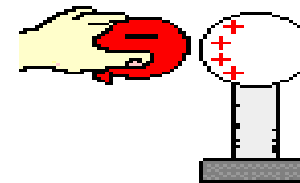
A - balloon induces e^- movement from the left side to the right side of the balloon.

Diagram iii.



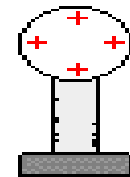
When touched, the e^- leave the sphere through the hand and enter "the ground."

Diagram iv.



The sphere is now charged positively, with the excess charge attracted to the balloon.

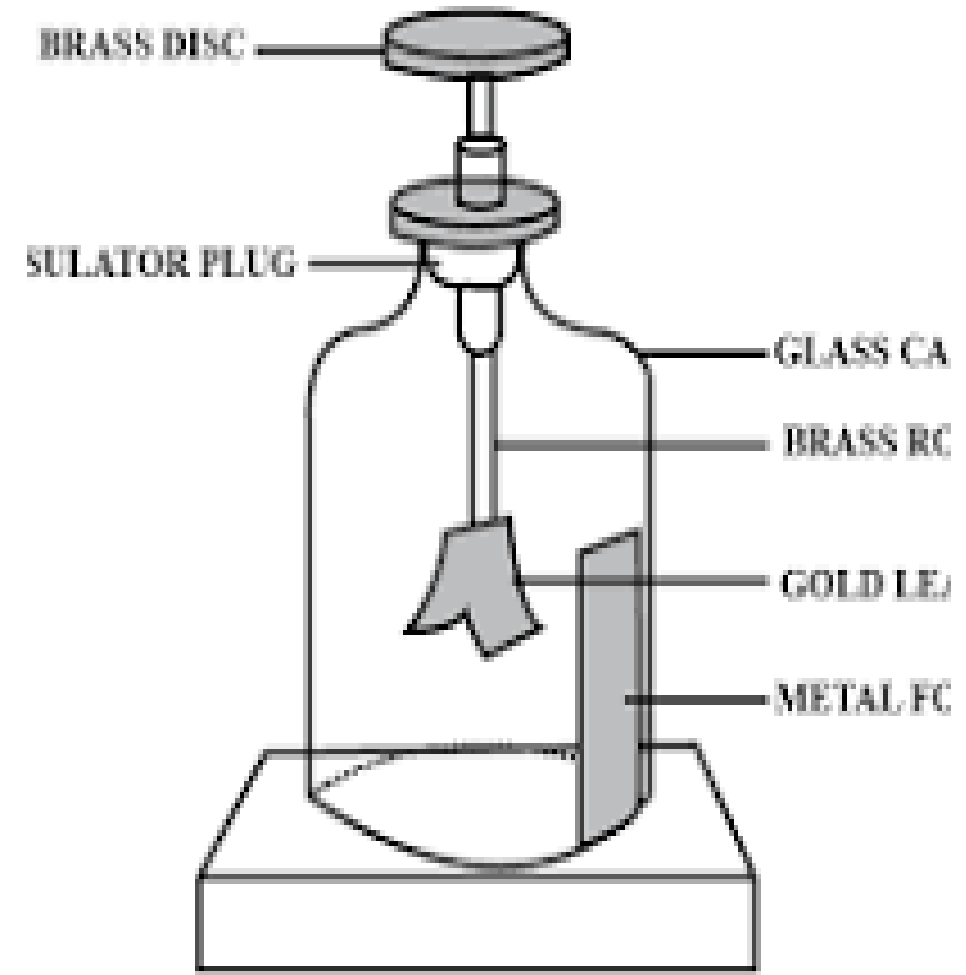
Diagram v.



The positive charge evenly distributes itself over the sphere.

The gold leaf electroscope

- This is an instrument for detecting the presence, determining the polarity and magnitude of charges.
- A metal disc is connected to a narrow metal plate and a thin piece of gold leaf is fixed to the plate.
- The whole of this part of the electroscope is insulated from the body of the instrument.
- A glass front prevents air draughts but allows you to watch the behavior of the leaf.



Gold leaf electroscope

The gold leaf electroscope

- When a charge is put on the disc at the top it spreads down to the plate and leaf. This means that both the leaf and plate will have the same charge. Similar charges repel each other and so the leaf rises away from the plate - the bigger the charge the more the leaf rises.

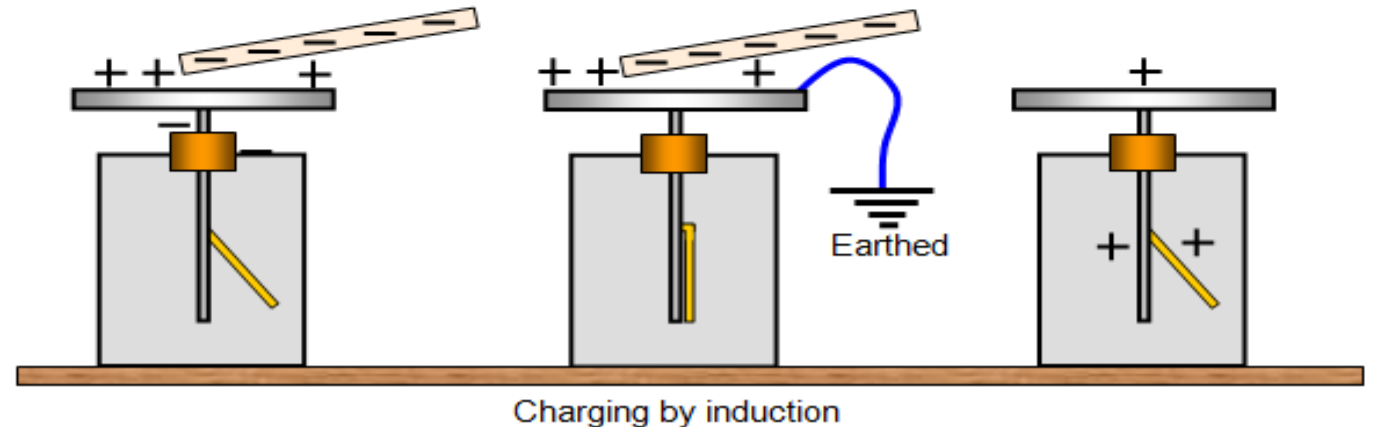
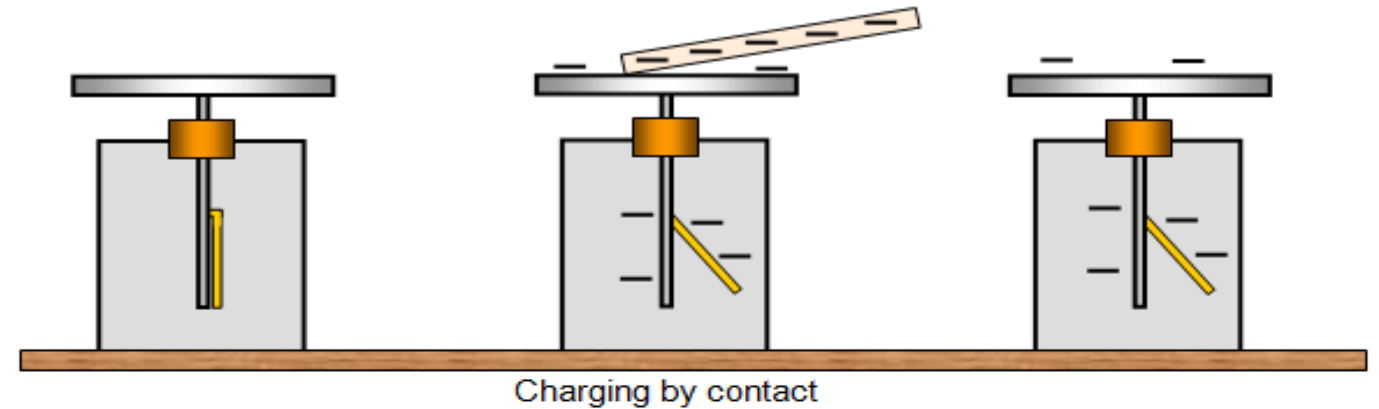
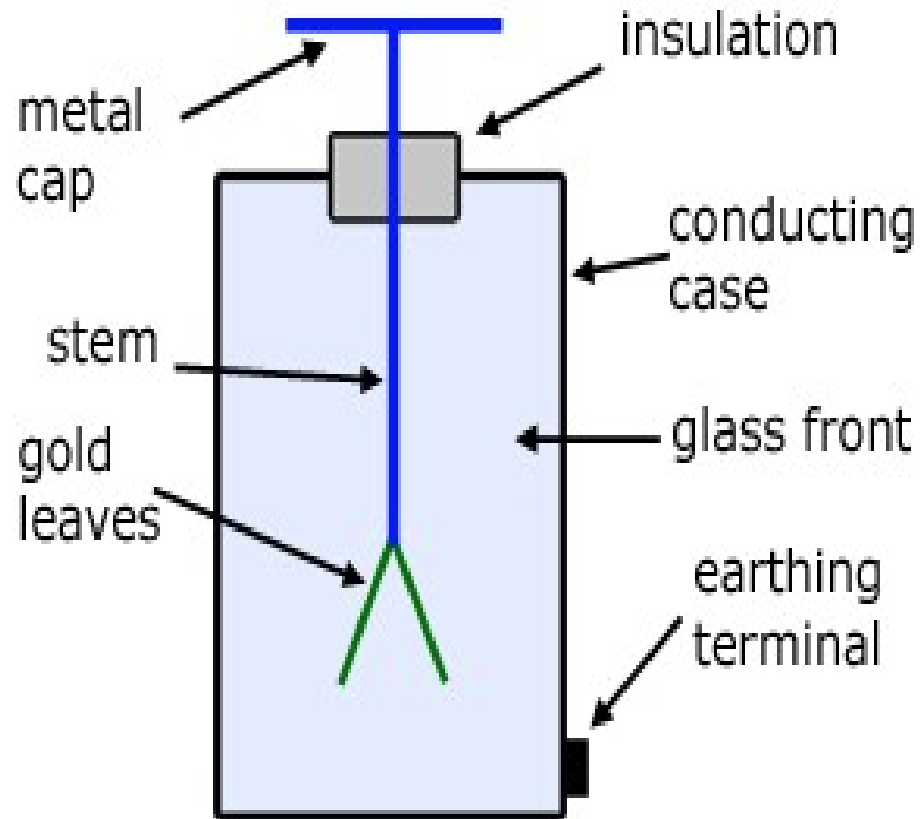
The leaf can be made to fall again by touching the disc - you have earthed the electroscope. An earth terminal prevents the case from becoming live. The electroscope can be charged in two ways

a) by contact - a charged rod is touched on the surface of the disc and some of the charge is transferred to the electroscope. This is not a very effective method of charging the electroscope.

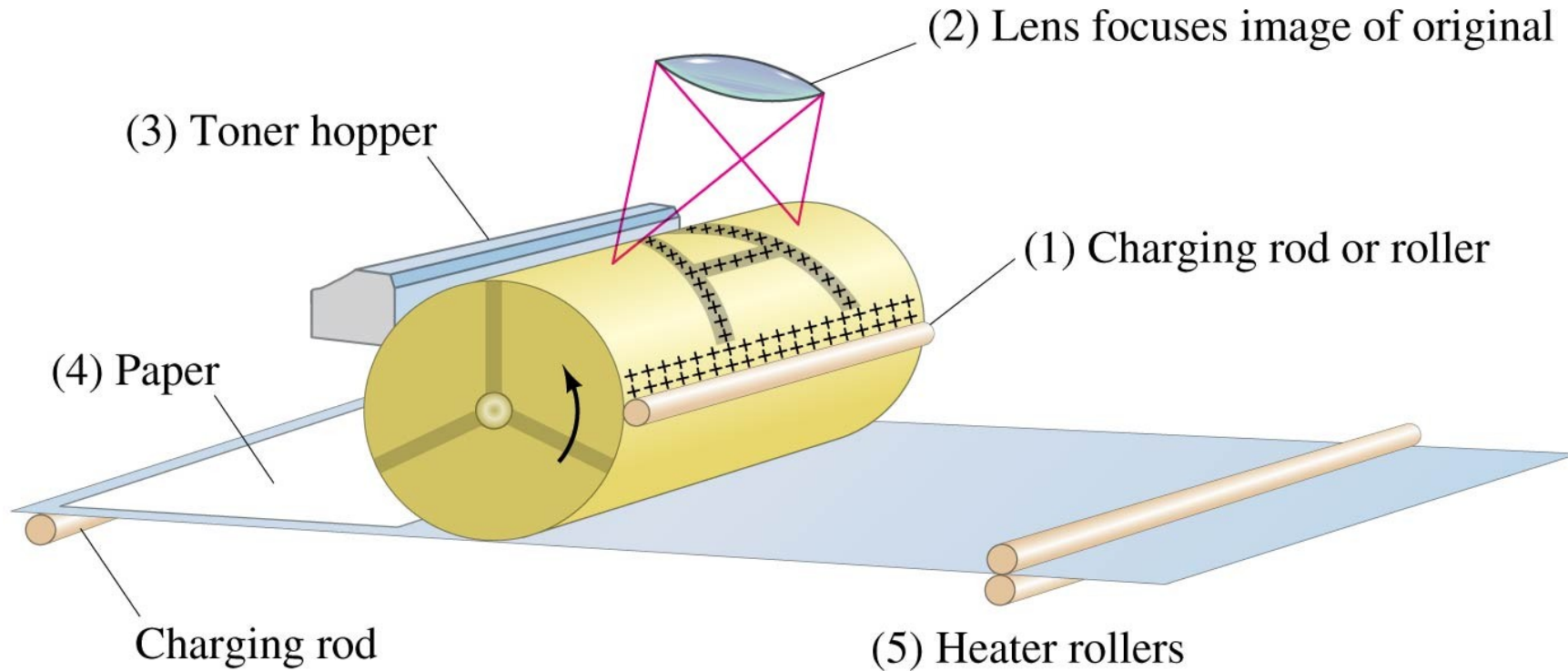
(b) by induction - a charged rod is brought up to the disc and then the electroscope is earthed, the rod is then removed. The two methods give the gold leaf opposite charges.

The gold leaf electroscope

The following diagrams show you how the charges spread over the plate and gold leaf in different conditions.

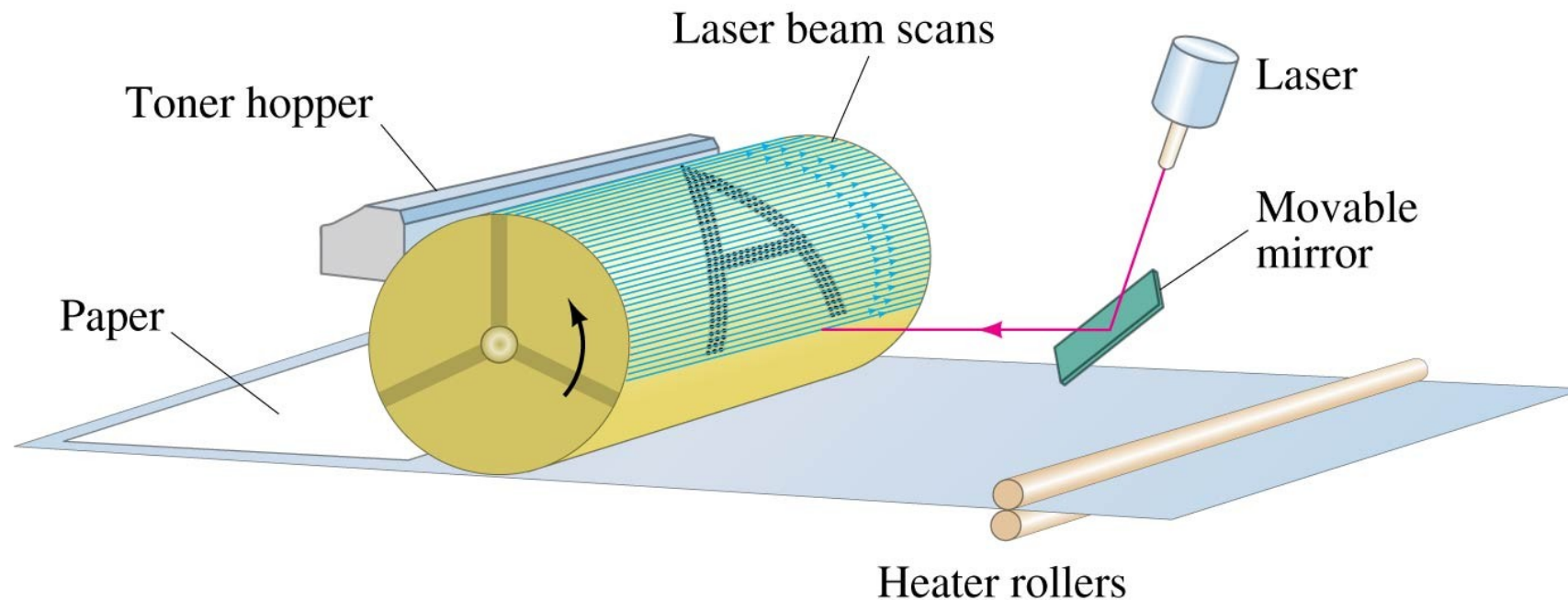


Photocopy Machines and Computer Printers Use Electrostatics



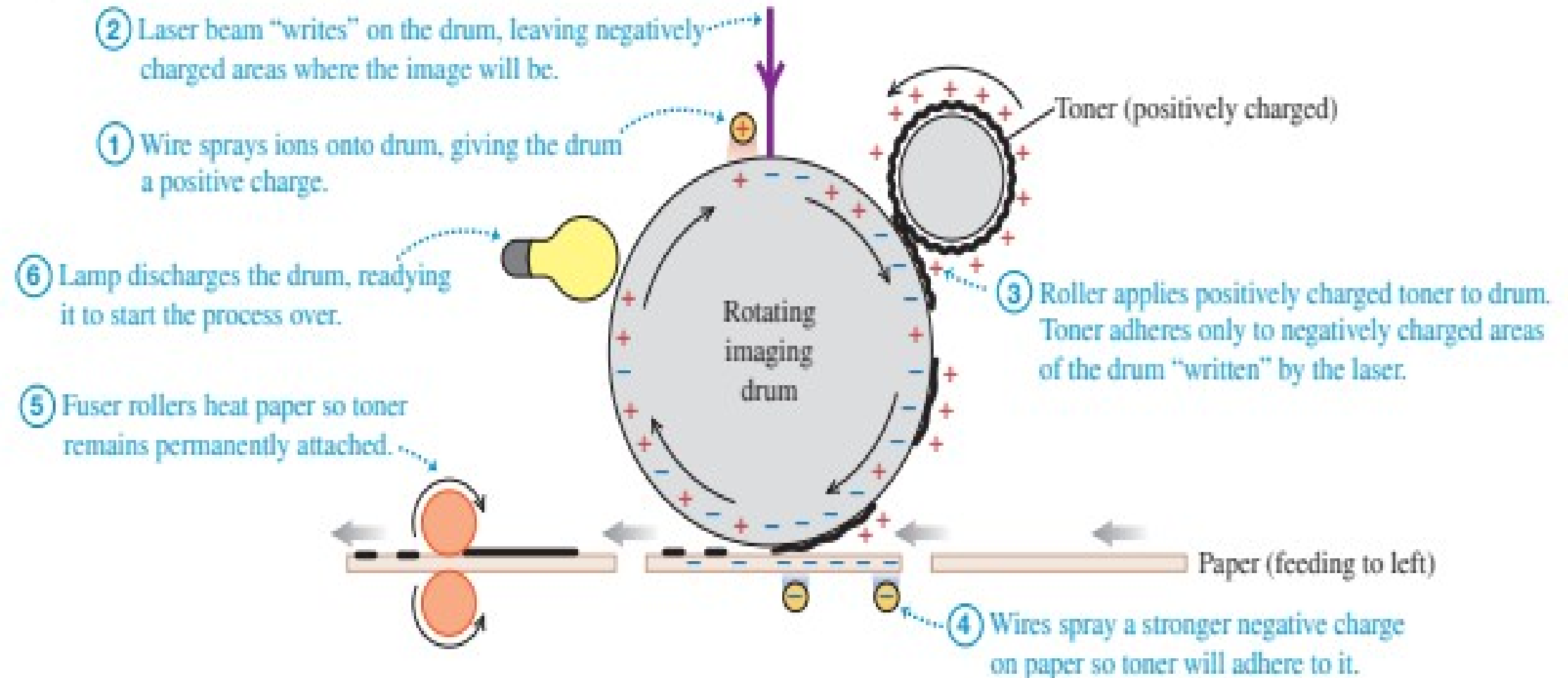
Photocopy Machines and Computer Printers Use Electrostatics

Laser printer is similar, except a computer controls the laser intensity to form the image on the drum



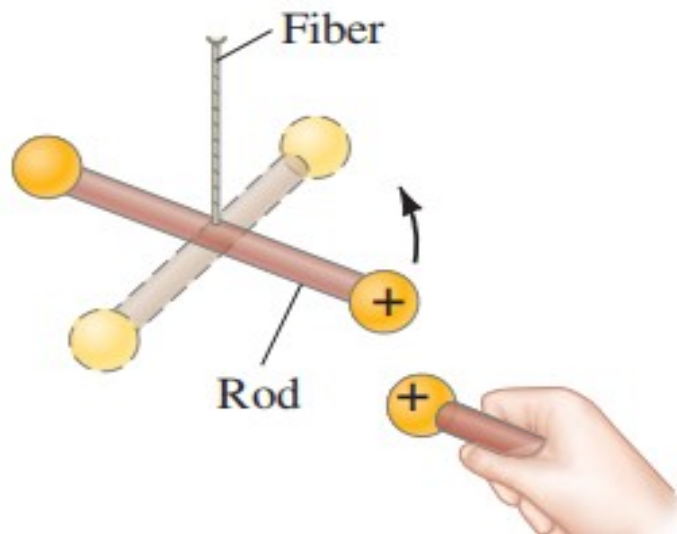
ELECTRIC CHARGE

Figure 21.2 Schematic diagram of the operation of a laser printer.

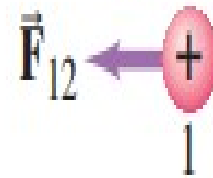


Electric force/coulomb's law

FIGURE 16–13 Coulomb's apparatus: when an external charged sphere is placed close to the charged one on the suspended bar, the bar rotates slightly. The suspending fiber resists the twisting motion, and the angle of twist is proportional to the force applied. With this apparatus, Coulomb investigated how the electric force varies as a function of the magnitude of the charges and of the distance between them.

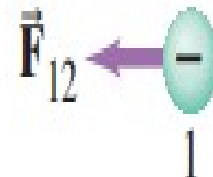
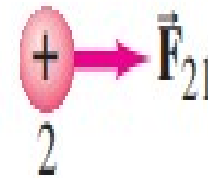


F_{12} = force on 1
due to 2

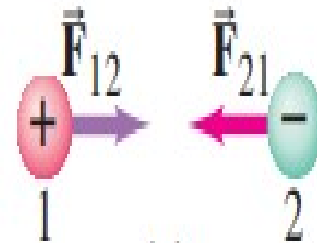
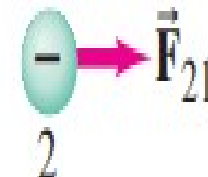


(a)

F_{21} = force on 2
due to 1

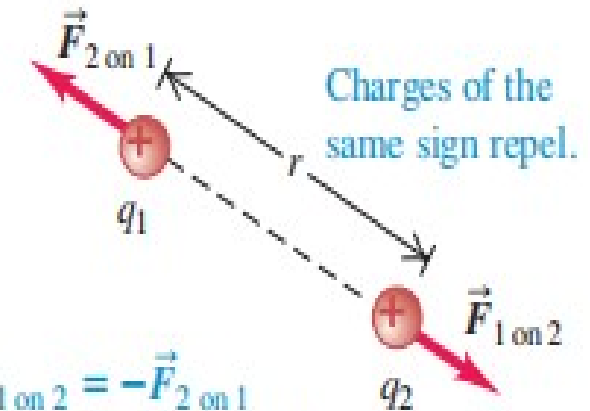


(b)



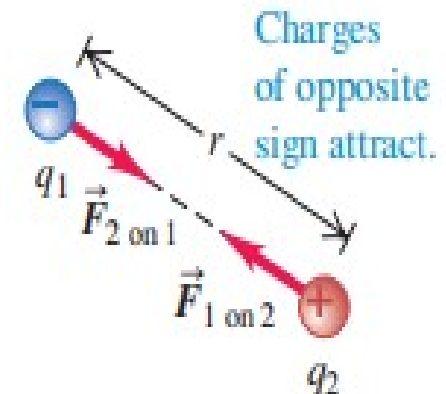
(c)

(b) Interactions between point charges



$$\vec{F}_{1 \text{ on } 2} = -\vec{F}_{2 \text{ on } 1}$$

$$F_{1 \text{ on } 2} = F_{2 \text{ on } 1} = k \frac{|q_1 q_2|}{r^2}$$



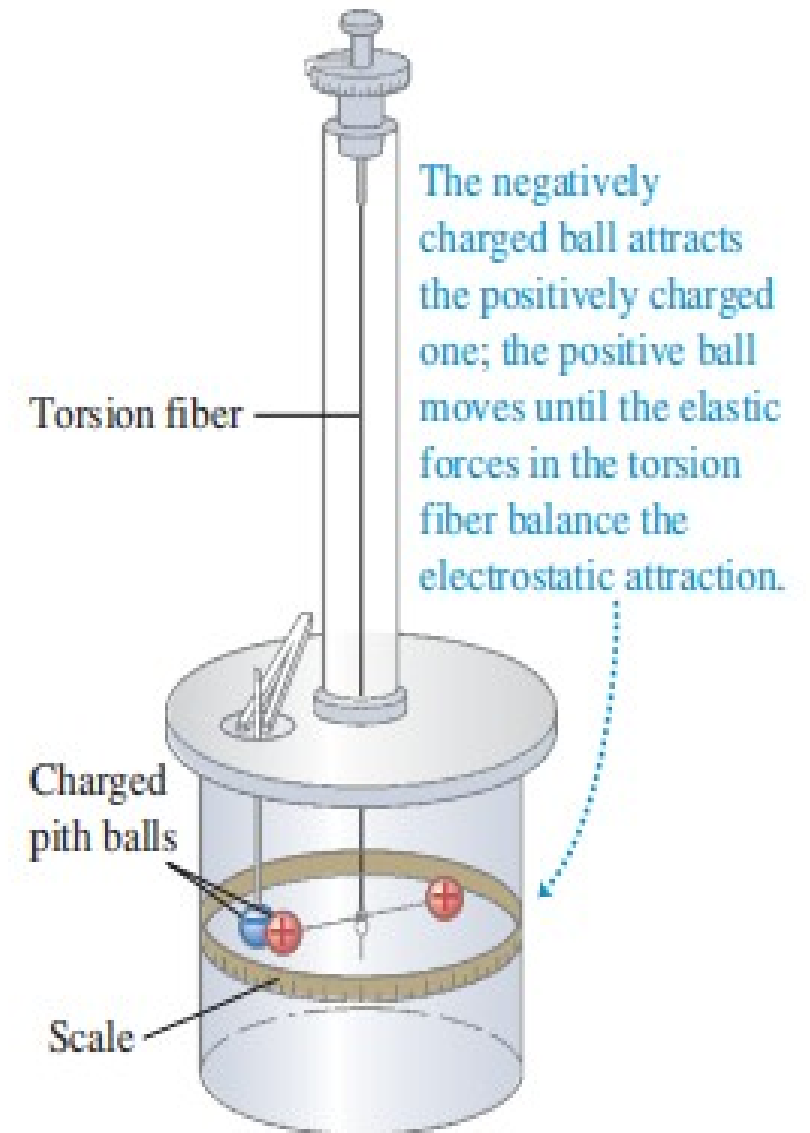
Electric Force

- Charles Coulomb (1736-1806) studied the magnitude of this force using a torsion balance.
- The **torsion balance** consists essentially of two small masses at different elevations that are supported at opposite ends of a beam. The latter is suspended from a wire that undergoes **torsion** because the masses are affected differently by the force of gravity.

Coulomb's law describes the force between two charges when they are at rest.

Additional forces come into play when charges are in motion. Coulomb's law gives the **electrostatic force**

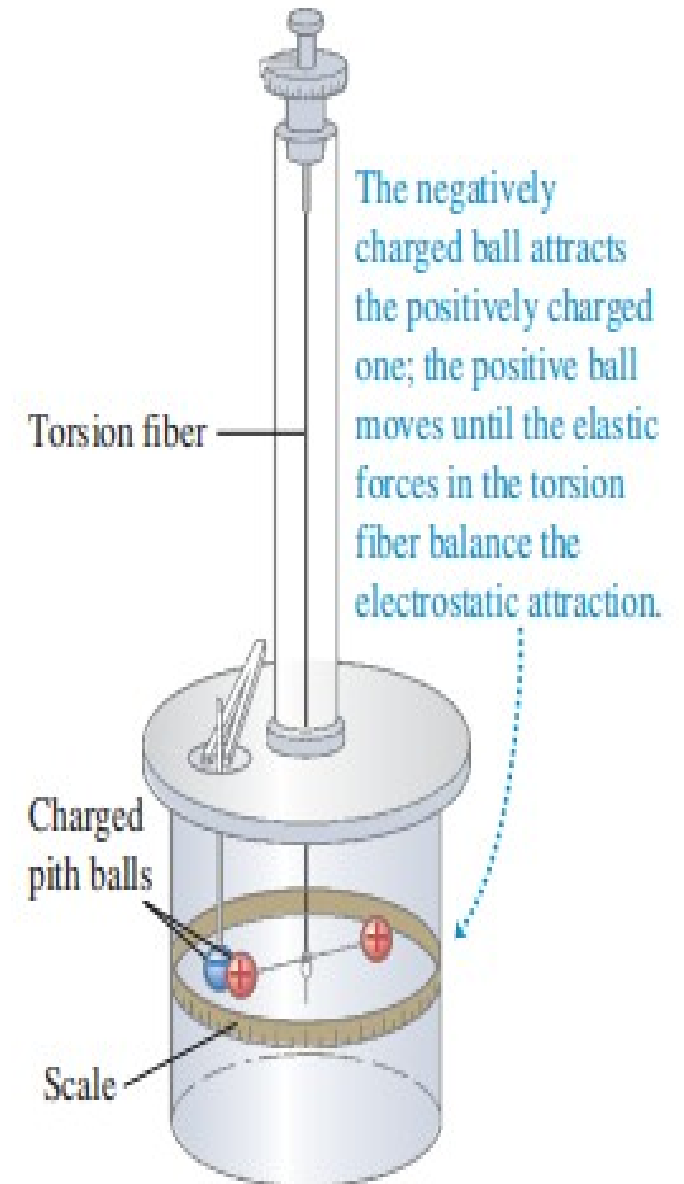
(a) A torsion balance of the type used by Coulomb to measure the electric force



Electric Force

- The experiment showed that the electrostatic force is directly proportional to the product of the charges on the metal spheres and inversely proportional to the square of the distance between them. **This was the basis of coulomb's law.**
- Charles Coulomb observed that when two electric charges are placed close to each other, *they experience a force.*
- He used a torsion balance to measure the repulsive and attractive forces between charged particles.

(a) A torsion balance of the type used by Coulomb to measure the electric force



Coulomb's law.

- ❑ **Coulomb's law states** that, the force of attraction or repulsion between the two electrically charged particles is directly proportional to the product of their charges and inversely proportional to the square of the distance between two charges, and acts along the line joining the two charges.

Coulomb's law looks a lot like the *law of universal gravitation*, which expresses the magnitude of the gravitational force a mass exerts on a mass. Both are **inverse square laws**

- ❑ This force of attraction or repulsion between two charges is also *depends on the medium in which charges are placed*.
- ❑ If we **increase** the distance between two-point charges, the force of attraction or repulsion present between them will **decrease**.
- ❑ In the similar way, if we **decrease** the distance between two-point charges, the force of attraction or repulsion present between them will **increase**.

Coulomb's law.

For the two-point charges, it depend directly on the product of the magnitude of the charges and inversely on the square of the distance between them:

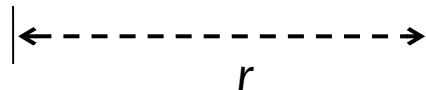
$$F_e \propto \frac{q_1 q_2}{r^2} \Rightarrow F_e = \frac{k q_1 q_2}{r^2}$$

$$k = \frac{1}{4\pi\epsilon_0} = 8.988 \times 10^9 \text{ Nm}^2 / \text{C}^2 \approx 9.00 \times 10^9 \text{ Nm}^2 / \text{C}^2$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 / \text{Nm}^2$$

Force on q_1 due to q_2

$$F_{12} = \frac{k q_1 q_2}{r^2}$$



Force on q_2 due to q_1

$$F_{21} = \frac{k q_1 q_2}{r^2}$$



The two Forces F_{12} and F_{21} are equal but opposite in direction

Coulomb's law.

- Where $\epsilon = \epsilon_0 \epsilon_r$
- ϵ_0 = permittivity of vacuum = **8.854×10^{-12} F/m**
- ϵ_r = relative permittivity of medium with respect to free space.
- For vacuum, the relative permittivity $\epsilon_r = 1$,
- Hence $\epsilon = \epsilon_0$
- Therefore, the force of attraction or repulsion between two electric charges that are placed in vacuum and medium is given by

$$k = \frac{1}{4\pi\epsilon}$$

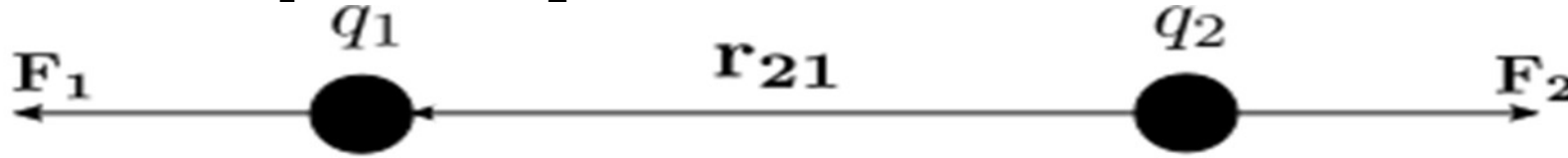
$$F = \frac{Q_1 Q_2}{4\pi\epsilon_0 \epsilon_r d^2} \quad \text{in medium}$$

$$F = \frac{Q_1 Q_2}{4\pi\epsilon_0 d^2} \quad \text{in vacuum}$$

-
-

Vector form of Coulomb's law

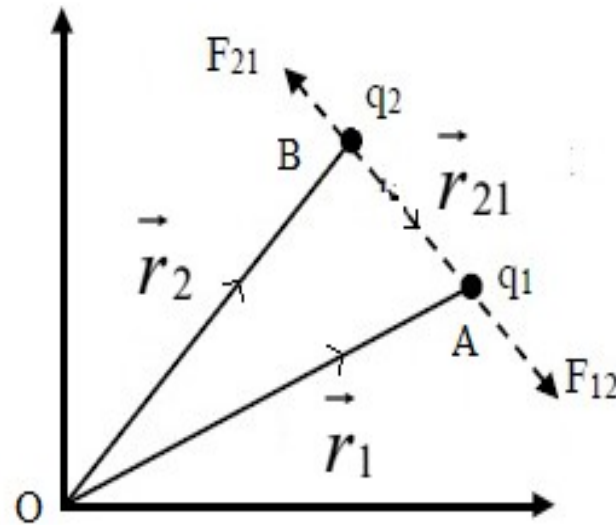
- Force on q_1 due to q_2



$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{21}^2} \hat{r}_{21}$$

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{21}^2} \frac{\vec{r}_{21}}{r_{21}}$$

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{21}^3} \vec{r}_{21} \dots$$



$$AB + OB = OA$$

$$AB = OA - OB$$

$$OA = r_1, OB = r_2, AB = r_{21}$$

$$r_{21} = r_1 - r_2$$

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{21}^3} (\vec{r}_2 - \vec{r}_1)$$

$$\vec{r}_{21} = \frac{r_{21}}{r_{21}} = \frac{r_1 - r_2}{r_1 - r_2}$$

$$\vec{r}_2 + \vec{r}_{21} = \vec{r}_1$$

$$\therefore \vec{r}_{21} = \vec{r}_1 - \vec{r}_2 \dots$$

Coulomb's law - vector form

If $\vec{F}_{21}$ is the force exerted on charge q_2 by charge q_1 (Fig. 1.3b),

$$\vec{F}_{21} = k \frac{q_1 q_2}{r_{12}^2} \hat{r}_{12}$$

where $\hat{r}_{12}$ is the unit vector from q_1 to q_2 .

If $\vec{F}_{12}$ is the force exerted on q_1 due to q_2 ,

$$\vec{F}_{12} = k \frac{q_1 q_2}{r_{21}^2} \hat{r}_{21}$$

where $\hat{r}_{21}$ is the unit vector from q_2 to q_1 .

[Both $\hat{r}_{21}$ and $\hat{r}_{12}$ have the same magnitude, and are oppositely directed]

$$\vec{F}_{12} = k \frac{q_1 q_2}{r_{12}^2} (-\hat{r}_{12})$$

$$\text{or } \vec{F}_{12} = -k \frac{q_1 q_2}{r_{12}^2} \hat{r}_{12}$$

$$\text{or } \vec{F}_{12} = -\vec{F}_{21}$$

So, the forces exerted by charges on each other are equal in magnitude and opposite in direction.

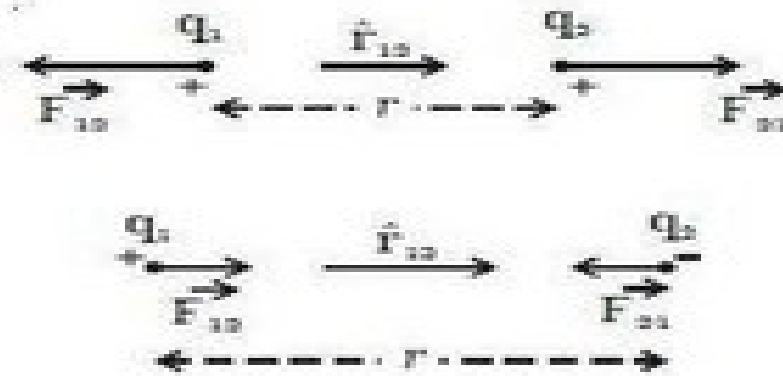


Fig 1.3b Coulomb's law in vector form

Limitations of Coulomb's law

- Coulomb's Law is applied to point charges. If charged bodies are of limited dimension such that they can't be regarded as a point charge, then Coulomb's Law is not valid. So, Coulomb's law is not valid for distance less than 10^{-15}
- Coulomb's law obeys inverse square law. It's only appropriate in those cases where inverse square law is obeyed.
- It is only applicable for point charges at rest. If a charge is in motion Coulomb's law of electrostatics is not valid
- It is difficult to apply Coulomb's law when the charges are in an arbitrary shape. Hence, we cannot determine the value of distance 'd' between the charges when they are in an arbitrary shape. Here, the distance, r cannot be resolved correctly as the centers of arbitrarily shaped charged bodies cannot be recognized exactly.

Conclusion from Coulomb's law

- Coulomb's law agrees with Newton's third law.

The two Forces F_{12} and F_{21} are equal but opposite in direction. “To every action there is equal and opposite reaction”

- Like charges repel and unlike charges attract.

Forces on q_2 due to q_1 will act away from each other if the charges are the same and will move towards each other if the charges are not the same.

Like charges repel



Opposites attract



Principle of superposition of charges

- It states that the total force on a given charge due to a few other charges is the vector sum of the individual forces acting on that charge by the other charges.
- Suppose there are q_1, q_2, q_3, q_4 and q_5 .
- The force on q_1 due to the other charges is given as
- If $\vec{F}_{12}$ then,

$$\vec{F}_{net} = \vec{F}_{12} + \vec{F}_{13} + \vec{F}_{14} + \vec{F}_{15}$$

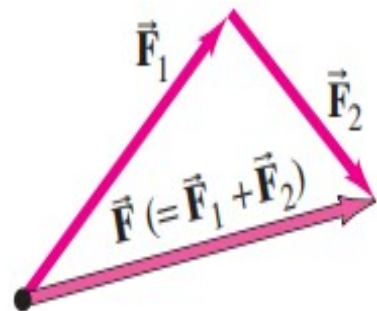
$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}$$

$$\vec{F}_{net} = \sum_{i=2}^n \vec{F}_{1i}$$

Solving Problems Involving Coulomb's Law and Vectors



(a) Two forces acting on an object.



(b) The total, or net, force is $\vec{F} = \vec{F}_1 + \vec{F}_2$ by the tail-to-tip method of adding vectors.

The electric force between charged particles at rest (sometimes referred to as the **electrostatic force** or as the **Coulomb force**) is, like all forces, a vector: it has both magnitude and direction. When several forces act on an object (call them $\vec{F}_1$, $\vec{F}_2$, etc.), the net force $\vec{F}_{\text{net}}$ on the object is the vector sum of all the forces acting on it:

$$\vec{F}_{\text{net}} = \vec{F}_1 + \vec{F}_2 + \cdots$$

$$\begin{aligned} F_{1x} &= F_1 \cos \theta_1 & F_{2x} &= F_2 \cos \theta_2 \\ F_{1y} &= F_1 \sin \theta_1 & F_{2y} &= -F_2 \sin \theta_2. \end{aligned}$$

We add up the x and y components separately to obtain the components of the resultant force $\vec{F}$, which are

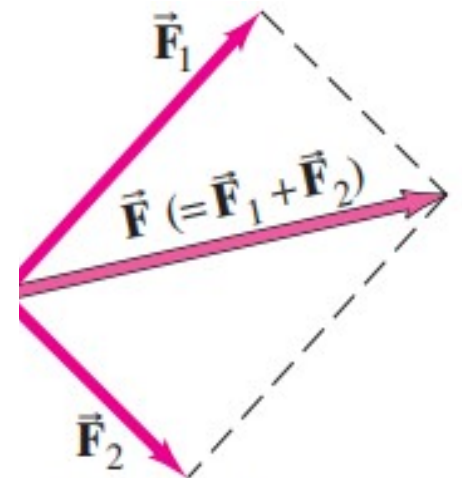
$$\begin{aligned} F_x &= F_{1x} + F_{2x} = F_1 \cos \theta_1 + F_2 \cos \theta_2, \\ F_y &= F_{1y} + F_{2y} = F_1 \sin \theta_1 - F_2 \sin \theta_2. \end{aligned}$$

The magnitude of the resultant (or *net*) force $\vec{F}$ is

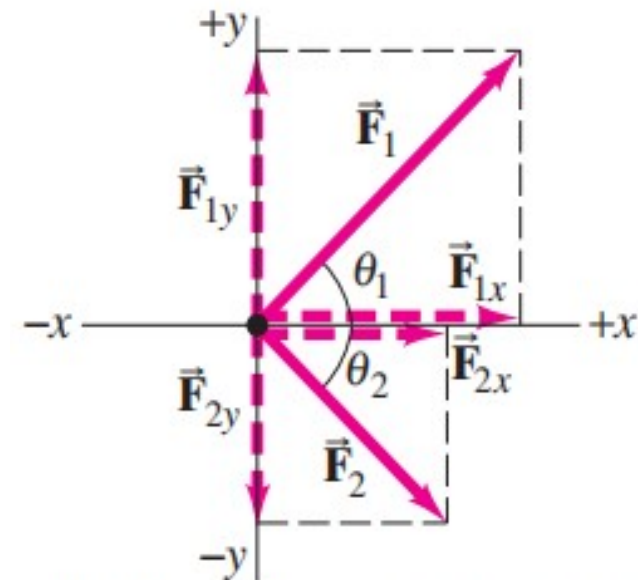
$$F = \sqrt{F_x^2 + F_y^2}.$$

The direction of $\vec{F}$ is specified by the angle θ that $\vec{F}$ makes with the x axis, which is given by

$$\tan \theta = \frac{F_y}{F_x}.$$



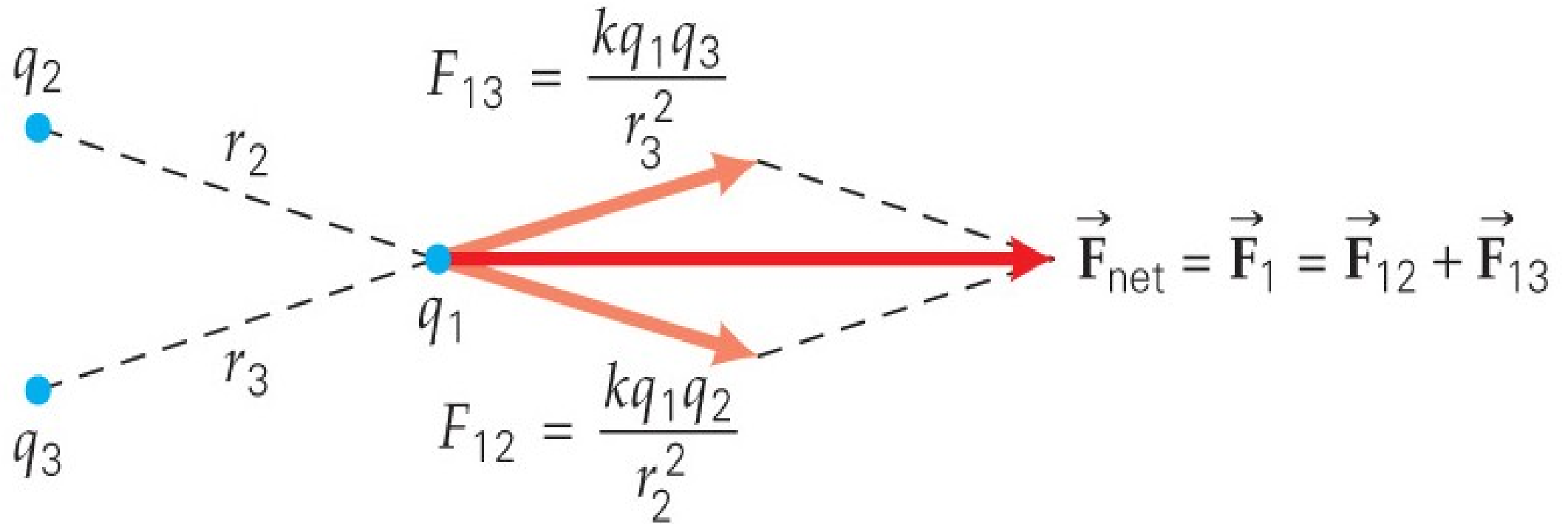
(c) $\vec{F} = \vec{F}_1 + \vec{F}_2$ by the parallelogram method.



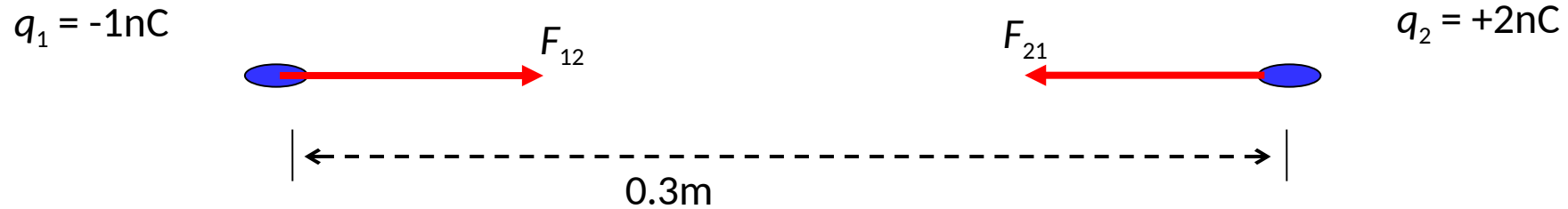
(d) $\vec{F}_1$ and $\vec{F}_2$ resolved into their x and y components.

Principle of superposition of charges

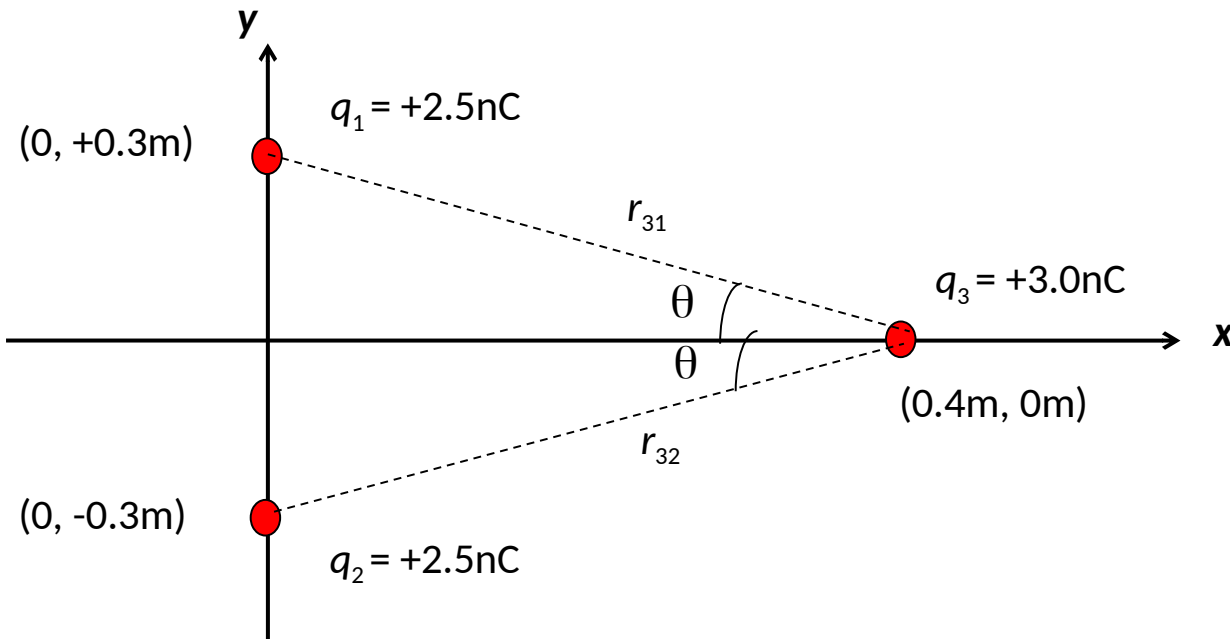
If there are **multiple point charges**, the force vectors must be added to get the net force.



EXAMPLE 1



(a) Two-point charges of -1.0nC and $+2.0\text{nC}$ are separated by a distance of 0.3m , what is the electric force on each particle?



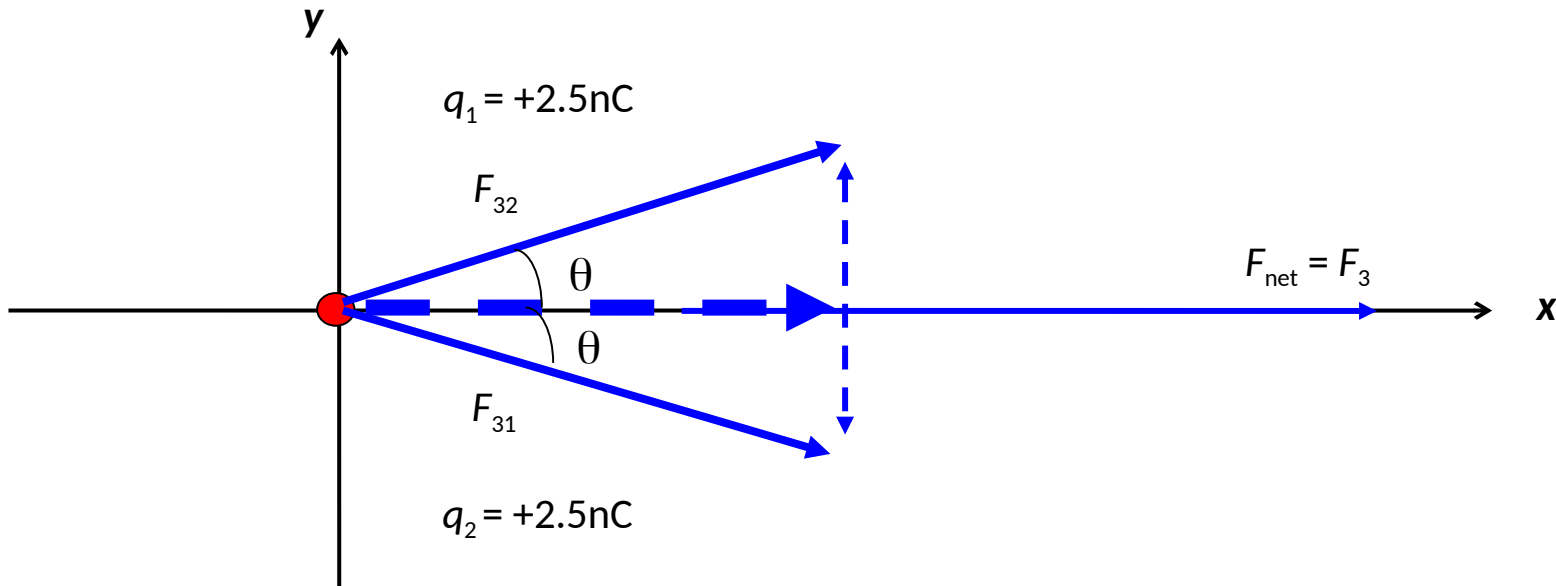
(b) What is the net electric force on q_3 ?

Solution:

(a)

$$\begin{aligned} F_{12} &= F_{21} = \frac{kq_1q_2}{r^2} \\ &= \frac{(9 \times 10^9 \text{ Nm}^2/\text{C}^2)(1 \times 10^{-9} \text{ C})(2 \times 10^{-9} \text{ C})}{(0.3\text{m})^2} \\ &= 0.2 \times 10^{-6} \text{ N} = 0.2 \mu\text{N} \end{aligned}$$

(b)



Solution:

(b) $r_{31} = r_{32} = 0.5\text{m}$

$$F_{32} = \frac{kq_2q_3}{r^2} = \frac{(9 \times 10^9 \text{ Nm}^2/\text{C}^2)(2.5 \times 10^{-9} \text{ C})(3.0 \times 10^{-9} \text{ C})}{(0.5\text{m})^2}$$
$$= 0.27 \mu\text{N}$$

Taking into account the direction of F_{31} and F_{32} is symmetry – then y – components cancel. Thus, F_3 (the net force on q_3) acts along the positive x-axis and has magnitude of

$$F_3 = F_{31} + F_{32} = 2F_{31} \quad \theta = \tan^{-1}\left(\frac{0.3\text{m}}{0.4\text{m}}\right) = 37^\circ$$

$$F_3 = 2F_{31} = 2F_{32} \cos \theta = 2(0.27 \mu\text{N})\cos 37^\circ = 0.43 \mu\text{N}$$

EXAMPLE 2

- (a) What is the magnitude of the repulsive electrostatic force between two protons in a nucleus? Taking the distance from center to center of these protons to be $3 \times 10^{-15} \text{m}$.
- b) If the protons were released from rest, how would the magnitude of their initial acceleration compare with that of the acceleration due to gravity on Earth's surface, g ?

Solution:

Given: $r = 3 \times 10^{-15} \text{m}$; $q_1 = q_2 = +1.6 \times 10^{-19} \text{C}$; $m_p = 1.67 \times 10^{-27} \text{kg}$

(a) Using Coulomb's Law;

$$F_e = \frac{kq_1q_2}{r^2} = \frac{(9 \times 10^9 \text{Nm}^2/\text{C}^2)(1.6 \times 10^{-19} \text{C})^2}{(3 \times 10^{-15} \text{m})^2} = 25.6 \text{N}$$

(b)

$$a = \frac{F_e}{m_p} = \frac{25.6 \text{N}}{1.67 \times 10^{-27} \text{kg}} = 1.53 \times 10^{28} \text{m/s}^2$$

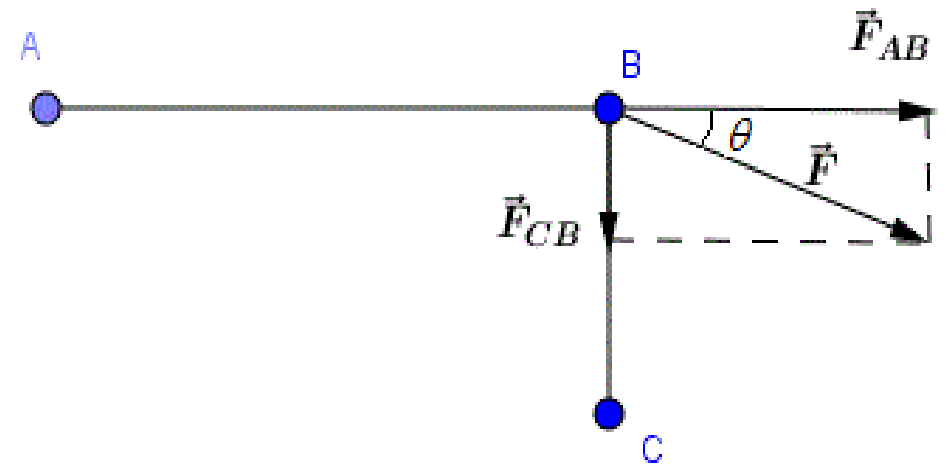
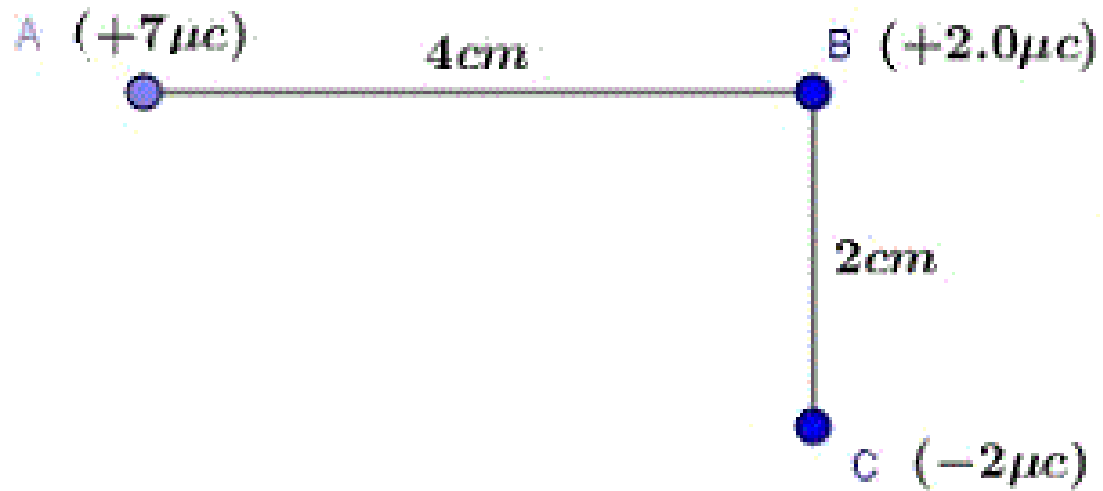
$$\frac{a}{g} = \frac{1.53 \times 10^{28} \text{m/s}^2}{9.8 \text{m/s}^2} = 1.56 \times 10^{27}$$

Example 3

- A positive charge q exerts a force of magnitude - 0.20 N on another charge - $2q$. Find the magnitude of each charge if the distance separating them is equal to 50 cm.
- The force that q exert on $2q$ is given by Coulomb's law:
- $F = k (q) (- 2q) / r^2$, $r = 0.5 \text{ m}$, $F = - 0.20 \text{ N}$,
- $- 0.2 = - 2 q^2 k / 0.5^2$
- $q^2 = 0.2 \times 0.5^2 / (2 k)$
- $q = \sqrt{ [(0.2 \times 0.5^2 / (2 \times 9 \times 10^9))] } = 1.66 \times 10^{-6} \text{ C}$
- $q = 1.66 \times 10^{-6} \text{ C}$, $-2 q = -3.32 \times 10^{-6} \text{ C}$

Example 4

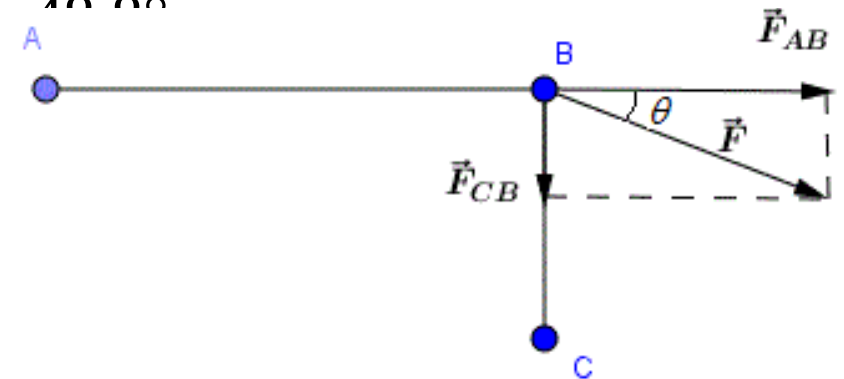
- What is the net force and its direction that the charges at the vertices A and C of the right triangle ABC exert on the charge in vertex B?



Solution 4

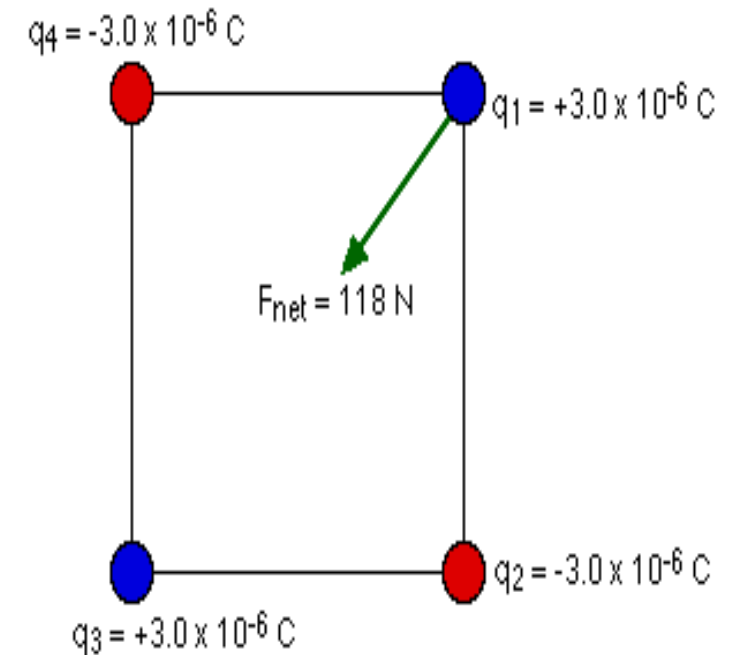
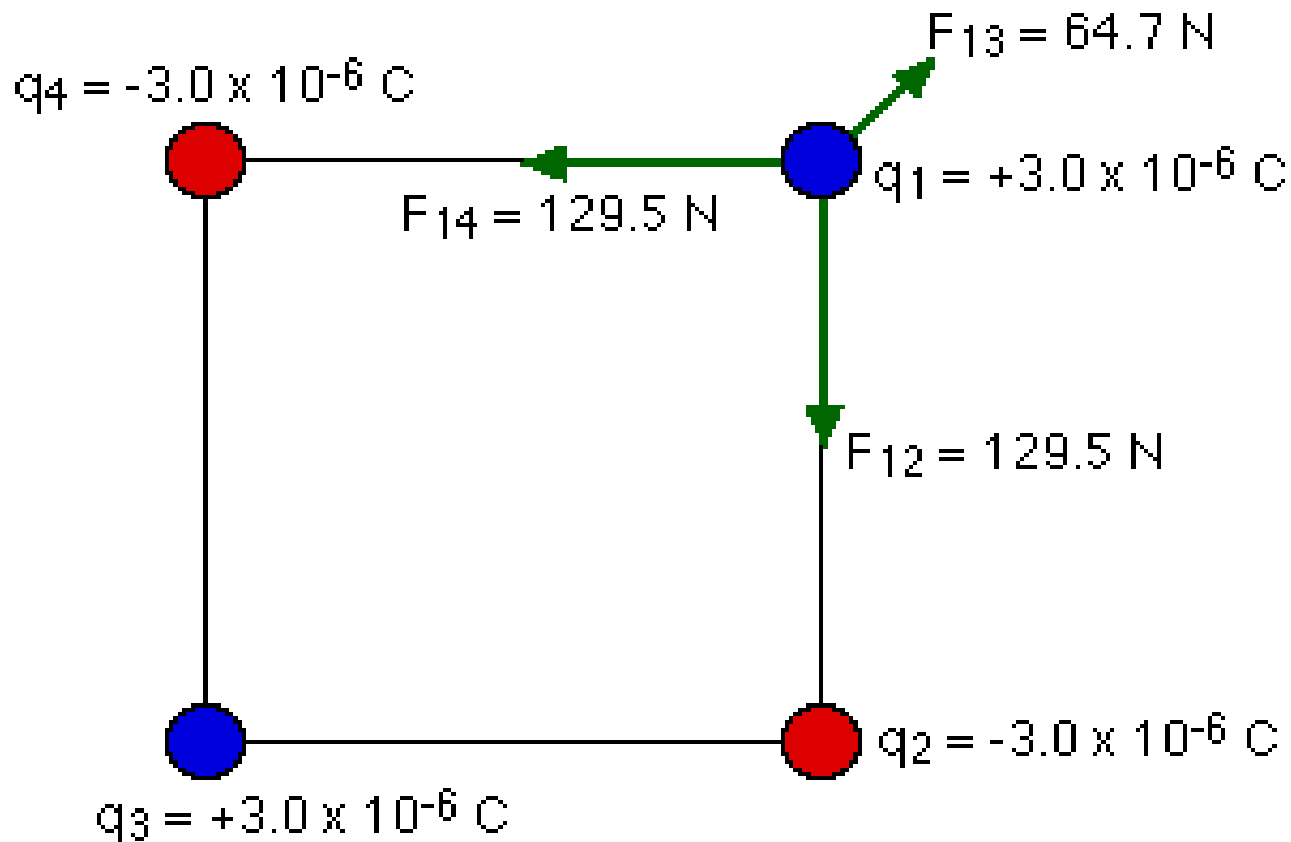
Let F_{BA} be the force of repulsion exerted by the charge at A on the charge at B and F_{BC} be the force exerted by the charge at point C on the charge at point B. The diagram below shows the direction of these two forces. We first use Coulomb's law ($F = k q_1 q_2 / r^2$) to find the magnitude of these two forces

- $|F_{BA}| = k (7 \times 10^{-6})(2 \times 10^{-6}) / (4 \times 10^{-2})^2 = 14 \times 10^{-12} k / (16 \times 10^{-4}) = 0.875 k \times 10^{-8}$
- $|F_{BC}| = k (2 \times 10^{-6})(2 \times 10^{-6}) / (2 \times 10^{-2})^2 = k \times 10^{-8}$
- We now use Pythagora's theorem to find the magnitude of the resultant force $F = F_{BA} + F_{BC}$ (a vector sum)
- $|F| = \sqrt{(|F_{BA}|^2 + |F_{BC}|^2)} = k \times 10^{-8} \sqrt{0.875^2 + 1^2} = 9.00 \times 10^9 \times 10^{-8} \sqrt{0.875^2 + 1^2} = 1.20 \times 10^2 \text{ N}$
- $\theta = \arctan(|F_{BC}| / |F_{BA}|) = \arctan(k \times 10^{-8} / 0.875 k \times 10^{-8}) = 48.01^\circ$
- solution of charges at vertices of right triangle



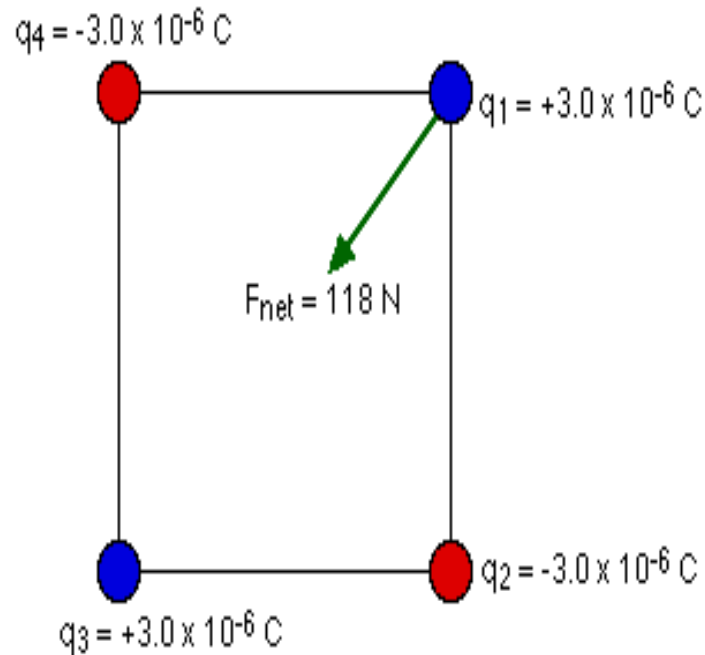
Example 5

Four charges are arranged in a square with sides of length 2.5 cm. The two charges in the top right and bottom left corners are $+3.0 \times 10^{-6} \text{ C}$. The charges in the other two corners are $-3.0 \times 10^{-6} \text{ C}$. What is the net force exerted on the charge in the top right corner by the other three charges?



Solution 5

To solve any problem like this, the simplest thing to do is to draw a good diagram showing the forces acting on the charge. You should also let your diagram handle your signs for you. Consider the forces exerted on the charge in the top right by the other three:



from charge 2 : $F_{12} = k q_1 q_2 / r^2 = (8.99 \times 10^9)(+3.0 \times 10^{-6})(-3.0 \times 10^{-6}) / (0.025)^2$
 $= -129.5 \text{ N} = +129.5 \text{ N}$ in the direction shown on the diagram

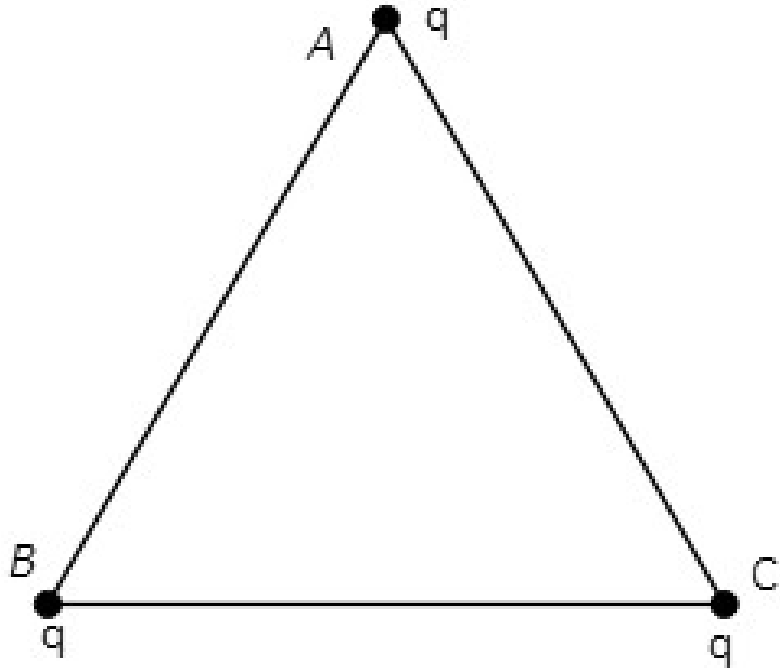
from charge 3 : (note that $r = 0.03536 \text{ m}$)

$F_{13} = k q_1 q_3 / r^2 = (8.99 \times 10^9)(+3.0 \times 10^{-6})(+3.0 \times 10^{-6}) / (0.03536)^2$
 $= +64.7 \text{ N} = +64.7 \text{ N}$ in the direction shown on the diagram

from charge 4 : $F_{14} = F_{12} = +129.5 \text{ N}$ in the direction shown on the diagram

Example 7

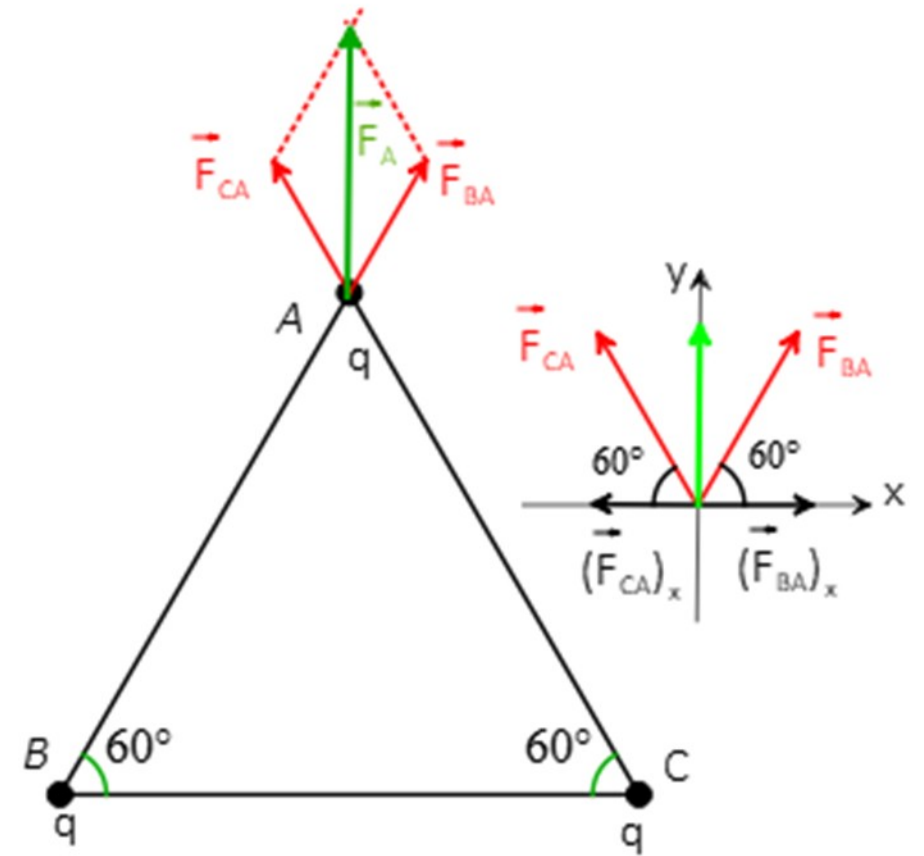
Three equal point charges are placed at the vertices of an equilateral triangle of side a . What is the magnitude and direction of the Coulomb force on the charge q at the point A? ($q=10\mu\text{C}$ and $a = \sqrt[4]{3}\text{ m}$).



Solution 7

first find the magnitudes of $\vec{F}_{BA}$ and $\vec{F}_{CA}$ using Coulomb's force law as below

$$\begin{aligned} F_{BA} &= k \frac{|q_B| |q_A|}{d^2} \\ &= k \frac{q^2}{(\sqrt{3})^2} \\ &= k \frac{q^2}{\sqrt{3}} \\ &= (9 \times 10^9) \frac{(10 \times 10^{-6})^2}{\sqrt{3}} \\ &= \frac{9}{\sqrt{3}} \times 10^{-1} \text{ N} \end{aligned}$$



Since the distance to q_A and the magnitudes of q_B and q_C are the same so $F_{BA} = F_{CA} = F$

Now find the direction of the electrostatic forces above using vector components. $\vec{F}_{BA}$ makes an angle of 60° with the $+x$ direction and $\vec{F}_{CA}$ an angle of 60° with the $-x$ direction. Thus, the above forces can be written in the following vector form

$$\vec{F}_{BA} = \underbrace{|\vec{F}_{BA}|}_F (\cos 60^\circ \hat{i} + \sin 60^\circ \hat{j})$$

$$\vec{F}_{CA} = \underbrace{|\vec{F}_{CA}|}_F (\cos 60^\circ (-\hat{i}) + \sin 60^\circ \hat{j})$$

The x -components will add up to zero which gives the x -component of the net force on the charge on the position A . The sum of the y -components also gives

$$F_{Ay} = F \sin 60^\circ + F \sin 60^\circ$$

$$= 2F \sin 60^\circ$$

$$= 2F \times \left(\frac{\sqrt{3}}{2} \right)$$

$$= \sqrt{3} F$$

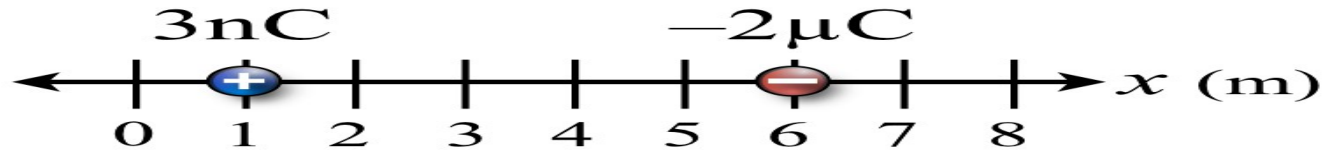
$$= 0.9 \text{ N}$$

$$= \sqrt{3} \times \frac{9}{\sqrt{3}} \times 10^{-1}$$

Therefore, the resultant Coulomb force on q_A directed upward and is written as $\vec{F}_A = 0.9 \hat{j}$.

Exercises

1. Consider a positive 3 nC charge and a negative $-2 \mu\text{C}$ charge sitting on the x axis as shown in the figure. Let's find the force on both charges, by allowing each one to be the target in turn.



2. Two charges 1 C and -3 C are kept at a distance of 3 m. Find the force of attraction between them.

3. Determine the electrical force of attraction between two balloons with separate charges of $+3.5 \times 10^{-8} \text{ C}$ and $-2.9 \times 10^{-8} \text{ C}$ when separated a distance of 0.65 m.

4. Calculate the electrostatic force and gravitational force between the proton and the electron in a hydrogen atom. They are separated by a distance of $5.3 \times 10^{-11} \text{ m}$. The magnitude of charges on the electron and proton are $1.6 \times 10^{-19} \text{ C}$. Mass of the electron is $m_e = 9.1 \times 10^{-31} \text{ kg}$ and mass of proton is $m_p = 1.6 \times 10^{-27} \text{ kg}$.

Exercise 9

- $+Q_1 = 10 \mu\text{C}$, $+Q_2 = 50 \mu\text{C}$ and Q_3 are separated as shown in the figure below. What is the electrostatic charge on particle 3 if the net electrostatic force on particle 2 is zero.

